Image Registration

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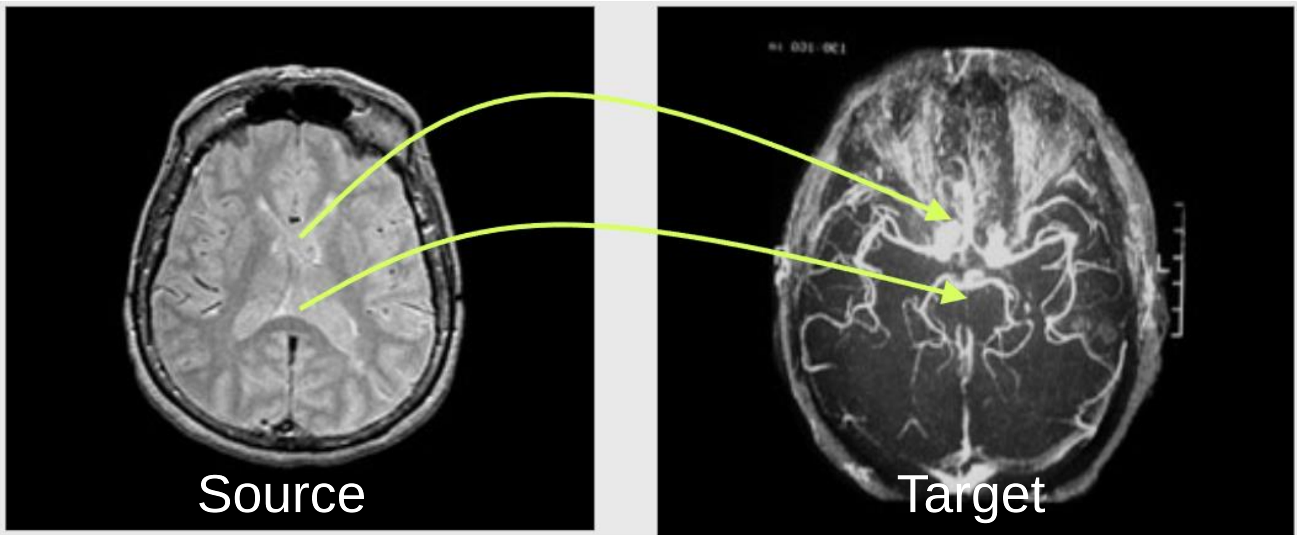
Why image registration?

- One patient was imaged
  - Today and one month before
  - After movement
  - On the CT and PET
  - Compared with
    - Another patient
    - The 'normalized / standard' image

Contents

Registration: definition

Spatial transform that maps points from one image to corresponding points in another image



- **Registration**
  - Optimise the spatial transformation between the source and reference (template) images
  - Re-sample according to the determined transformation parameters

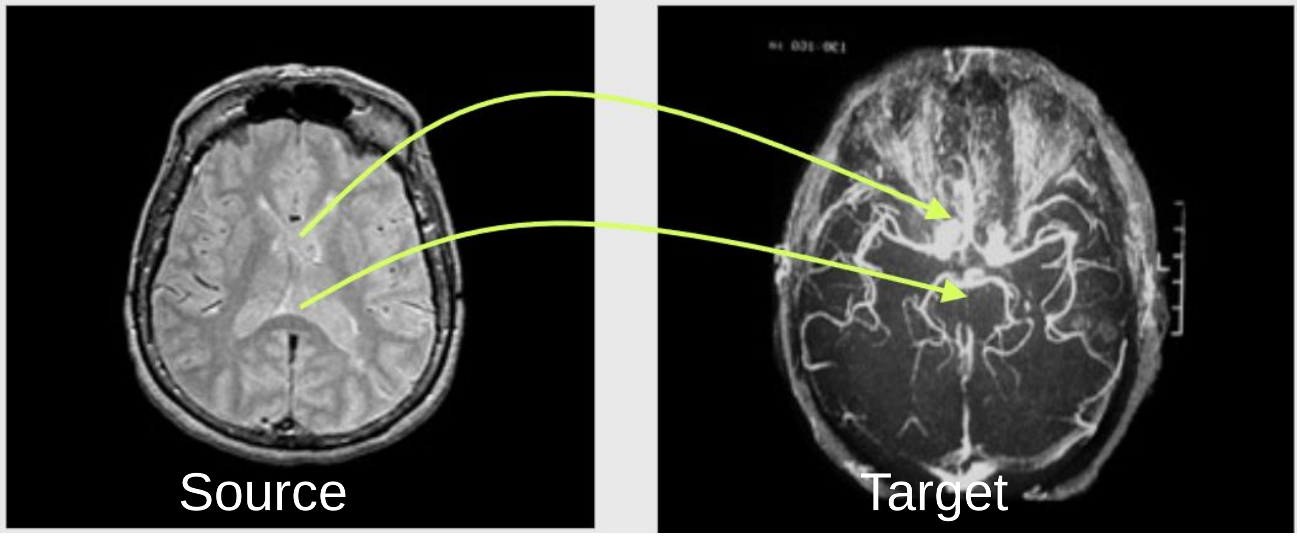
Registration: definitions

Given a set of matched feature points:

$$\{x_i, x_i'\}$$

Points in the source image

Points in the target image



And a transformation:

$$x_i' = f(x_i; p)$$

Transformation function

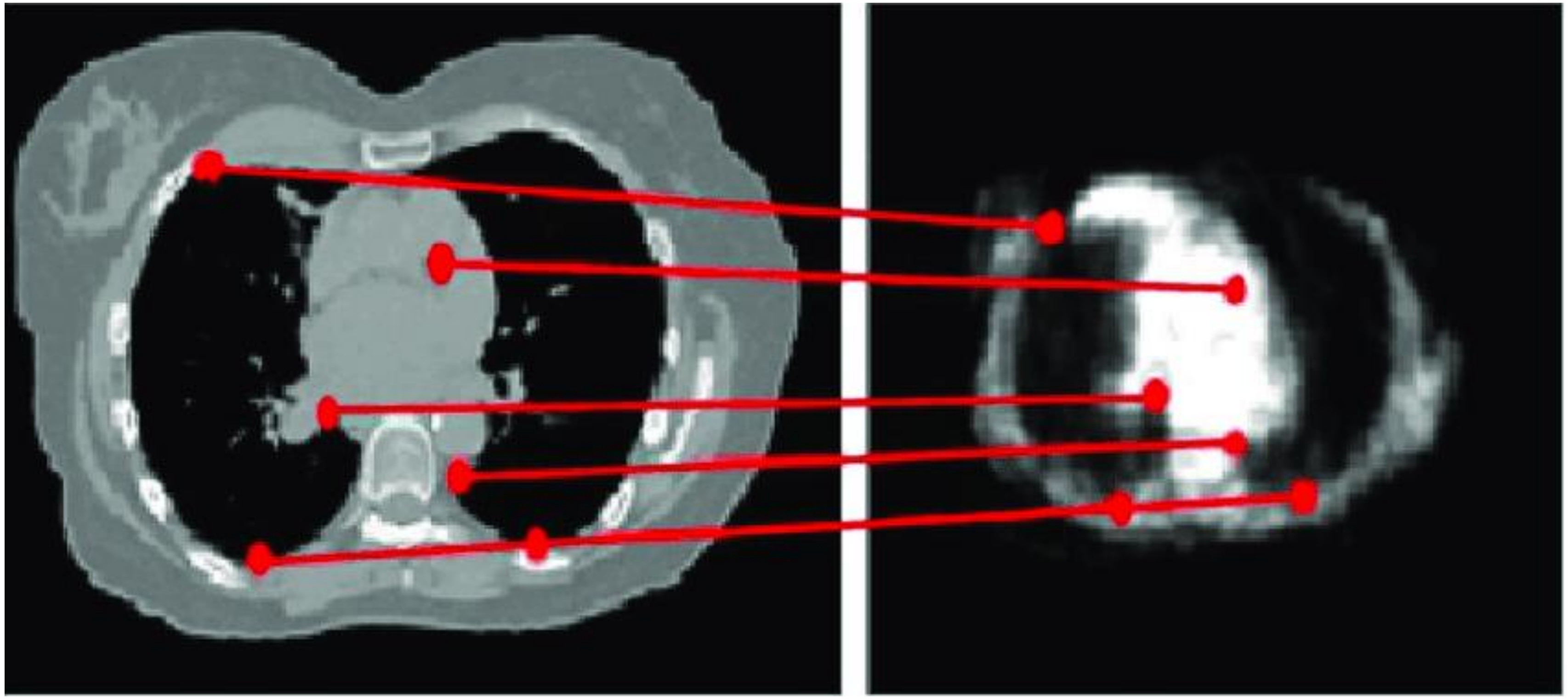
parameters

Goals

- To define the transformation function f
- To optimize the parameters p

Registration Basis

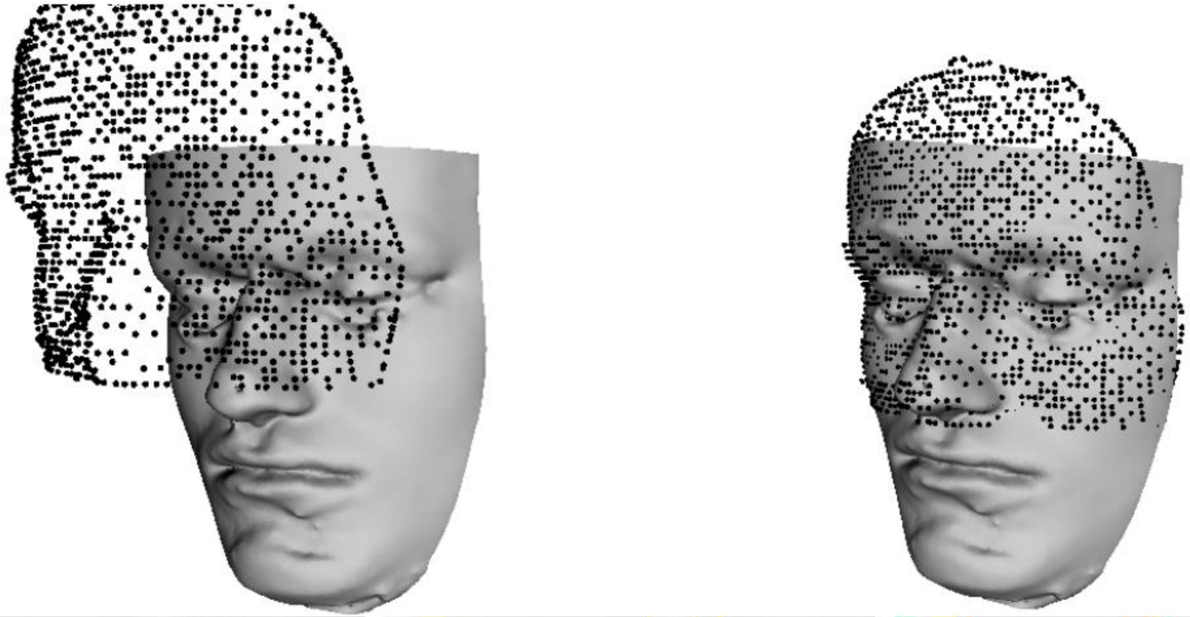
- Point-based methods
- Surface-based methods
- Intensity-based methods

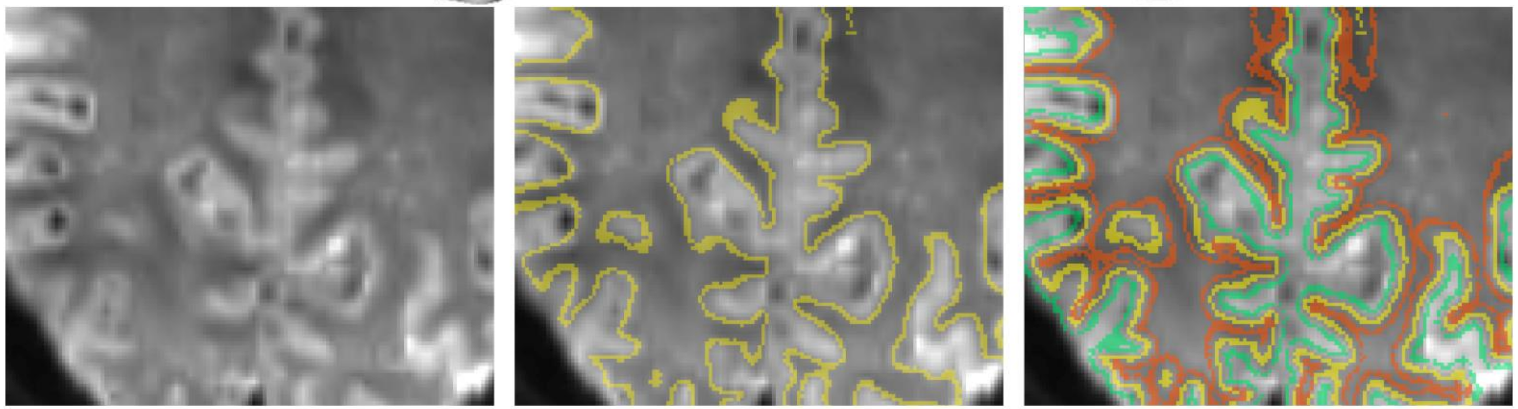


Landmark based registration between CT-PET

Registration Basis

- Point-based methods
- Surface-based methods
- Intensity-based methods

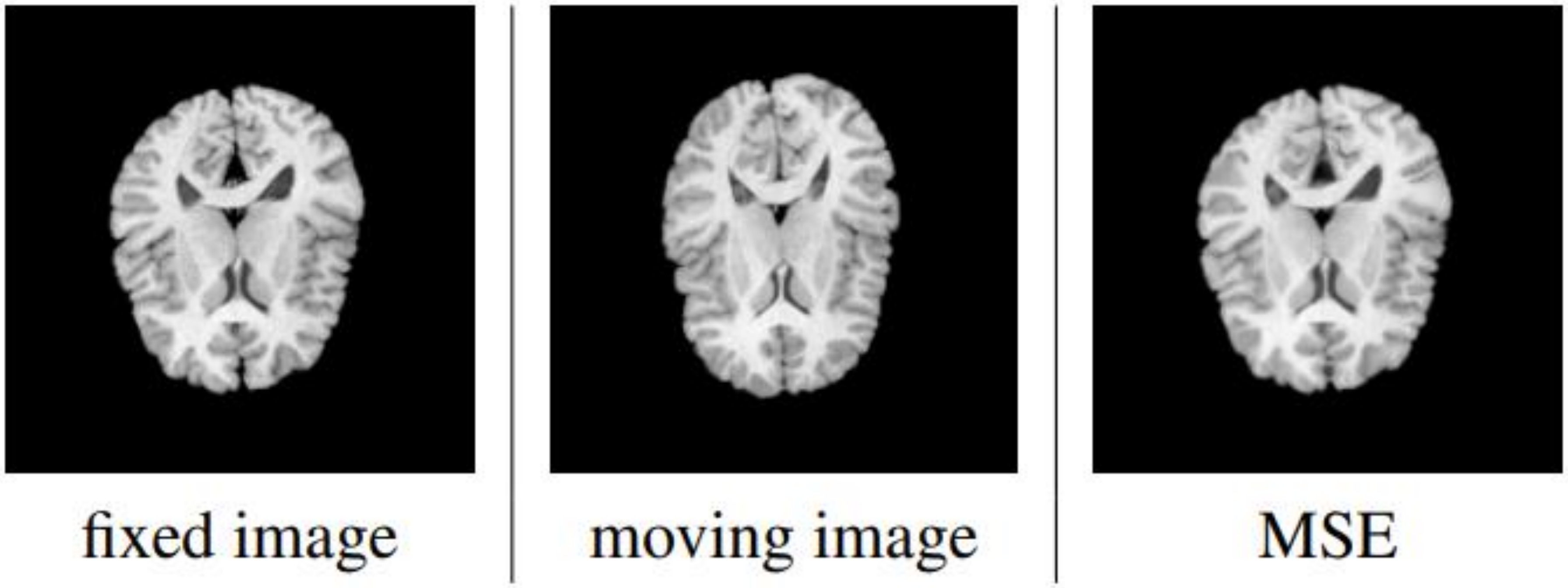




Original WM/GM cortex mesh WM (red), GM (green)

Registration Basis

- Point-based methods
- Surface-based methods
- **Intensity-based methods**



Registered

Registration Pipeline

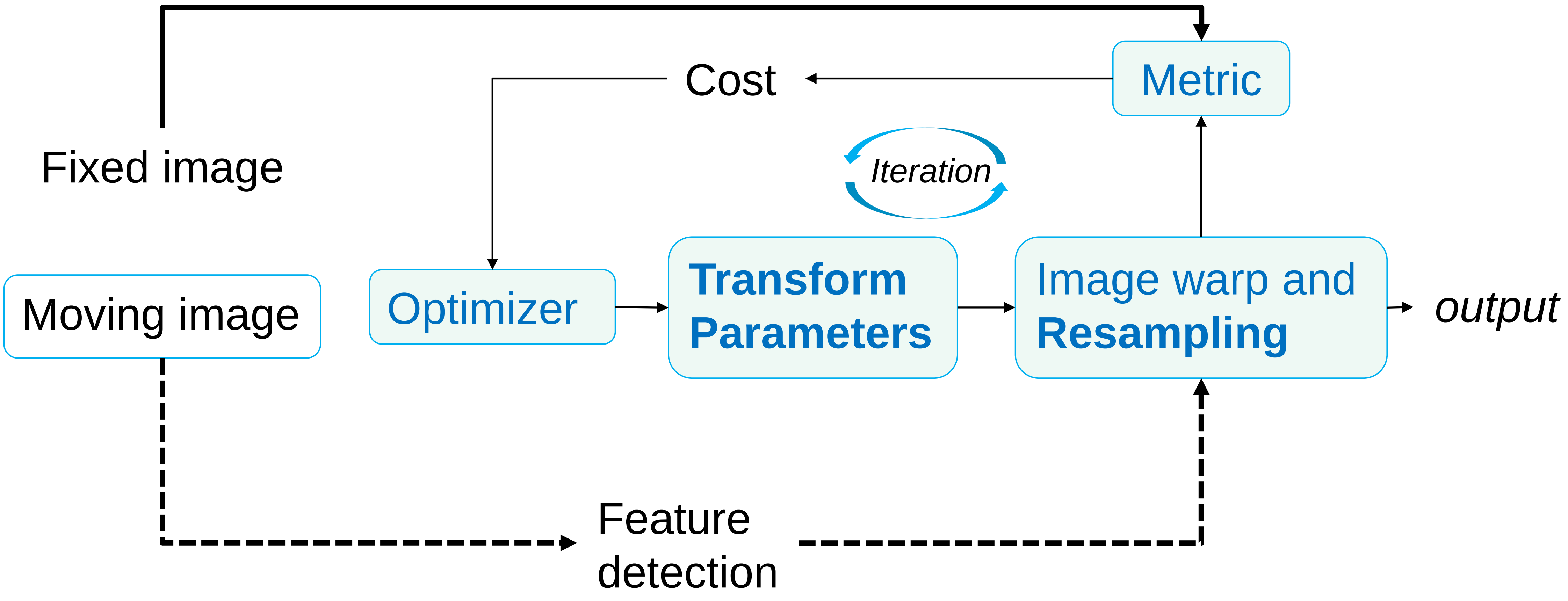
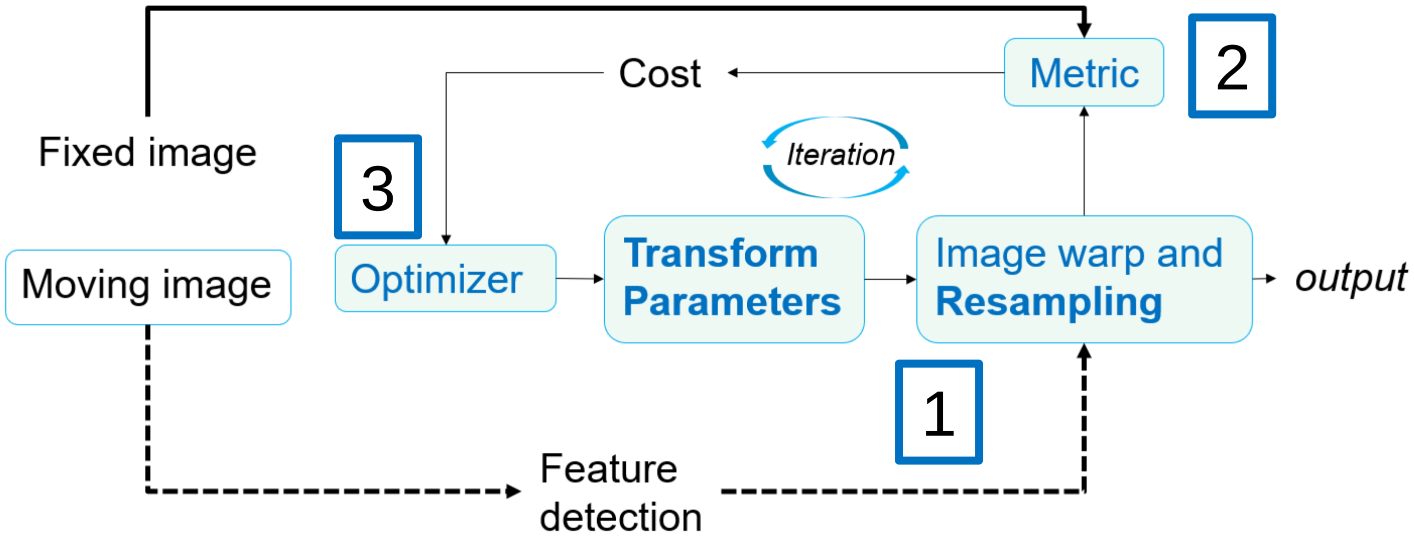
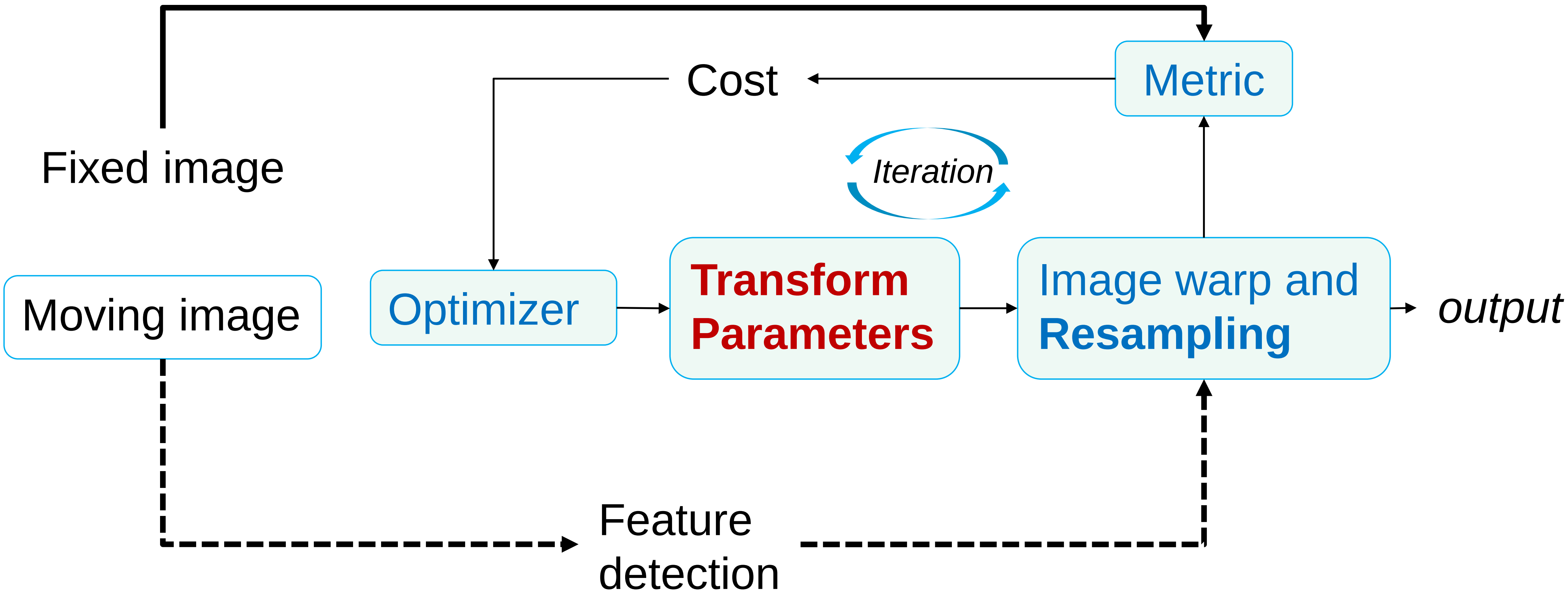


Image Registration Basic Steps

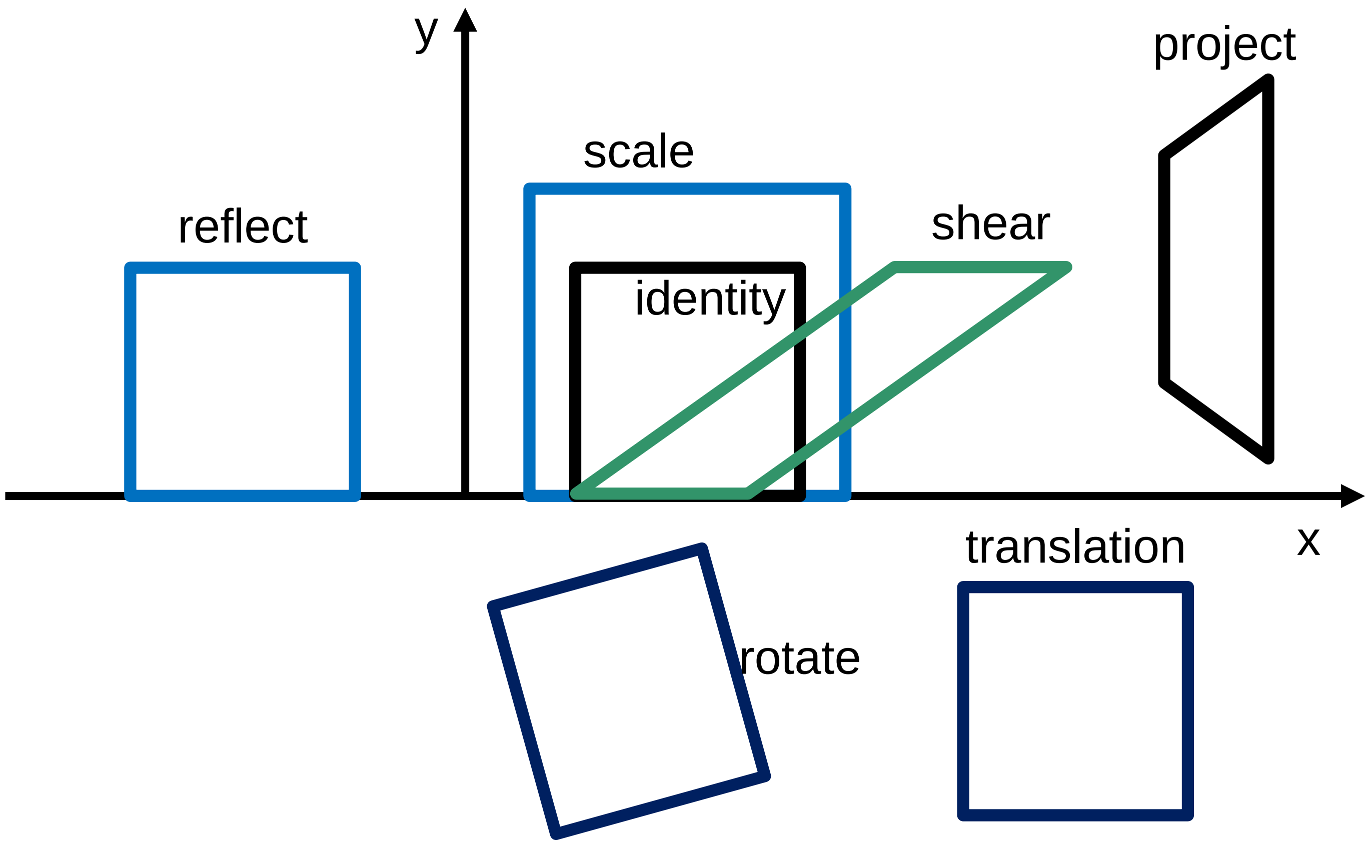
1. Define class of transformations (rigid body, affine, nonlinear)
2. Define similarity index / target metric (homologous features, intensity)
3. Optimize transformation parameters to maximize similarity index (a.k.a minimize the cost function)



Registration Pipeline



Linear Transformation



scale rotation translation

$$T = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

shear

Affine Transform

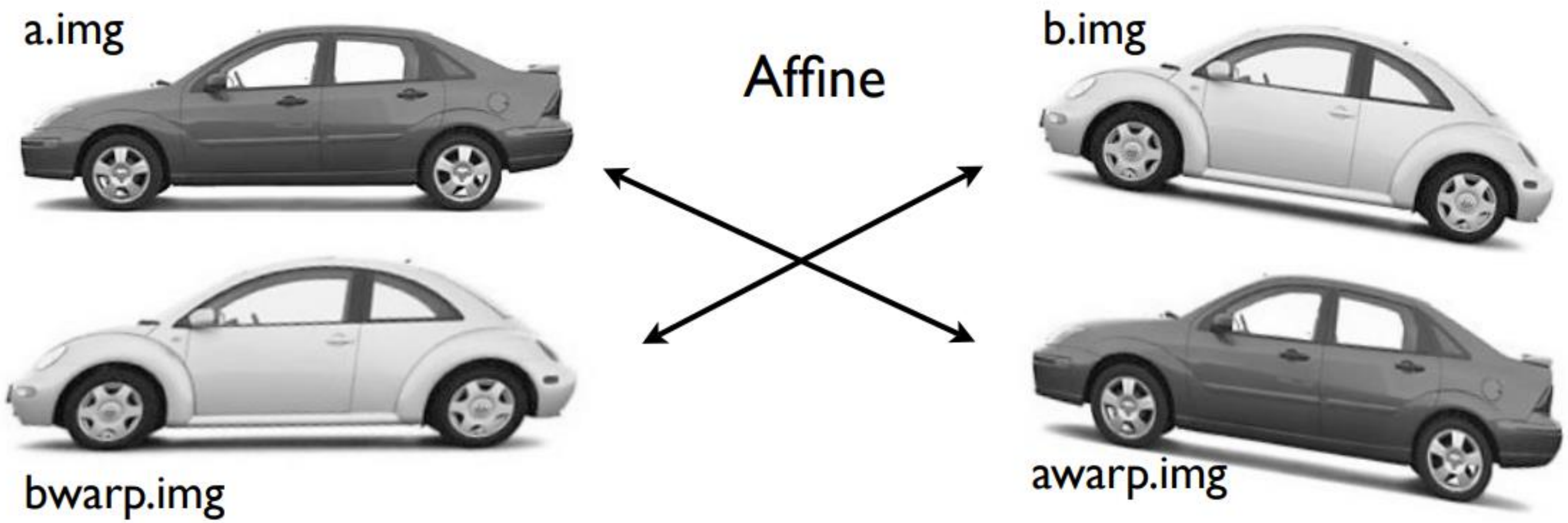
- Translation
- Rotation
- Scaling
- Shearing

rigid

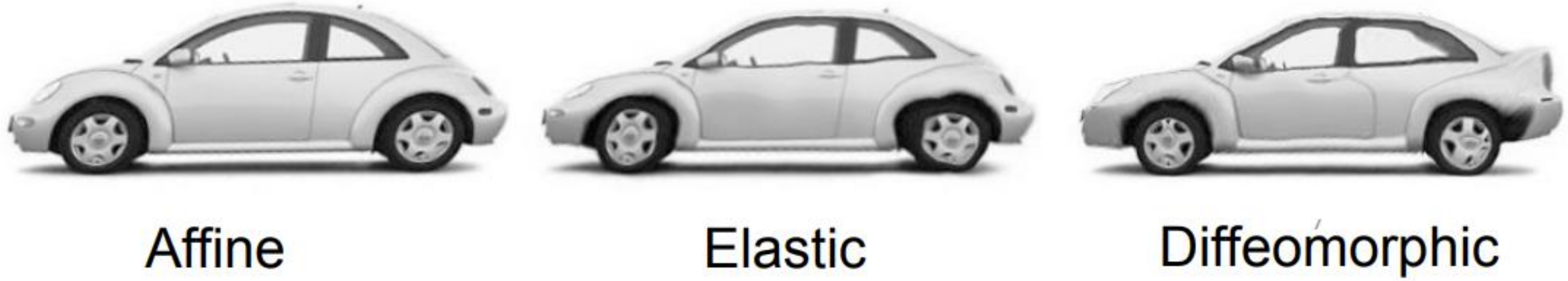
Affine

Linear and Non-linear Transformations

- Affine transformation (12 degrees of freedom)

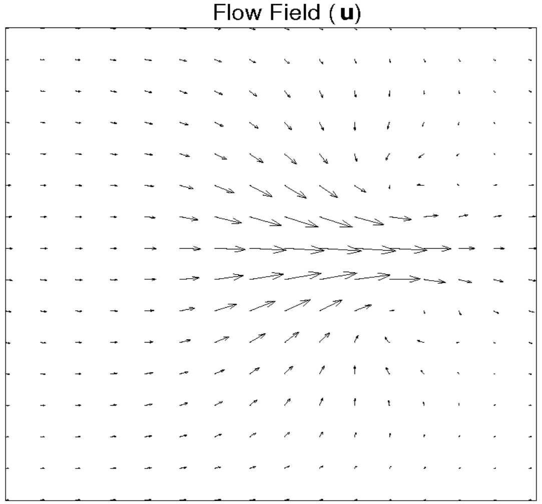


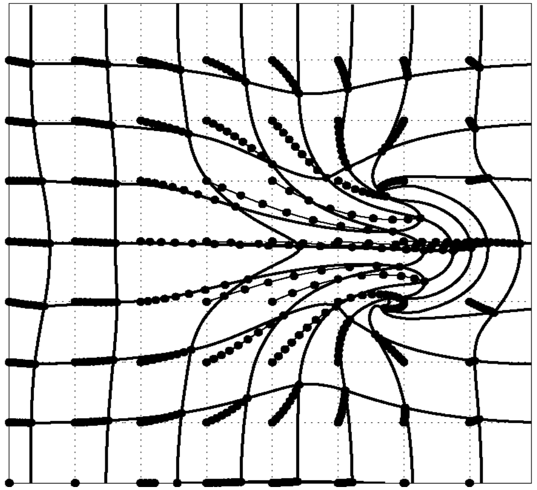
- Increasing degrees of freedom

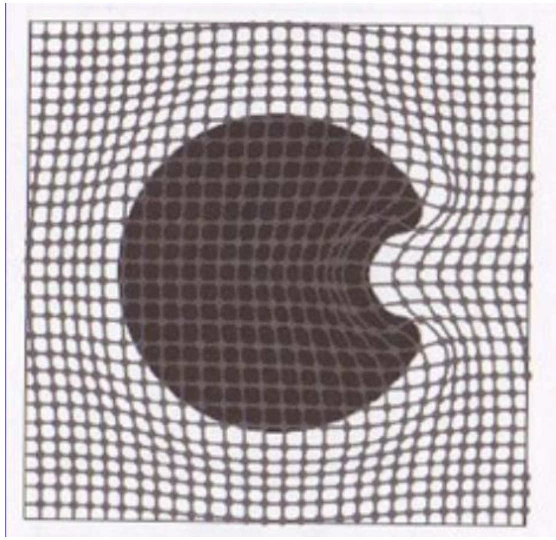


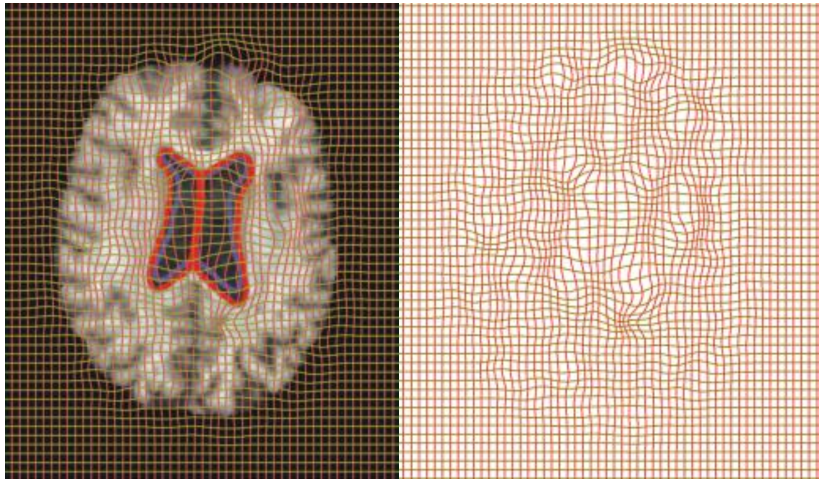
Non-linear Transformation

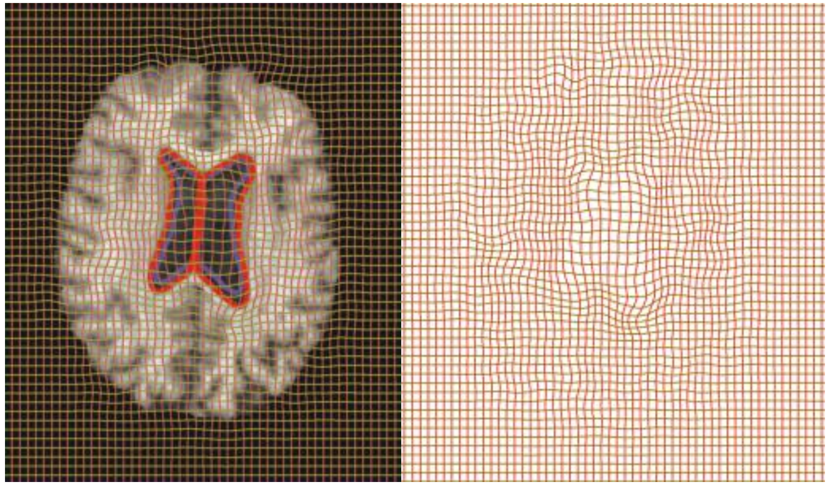
Deformation Field



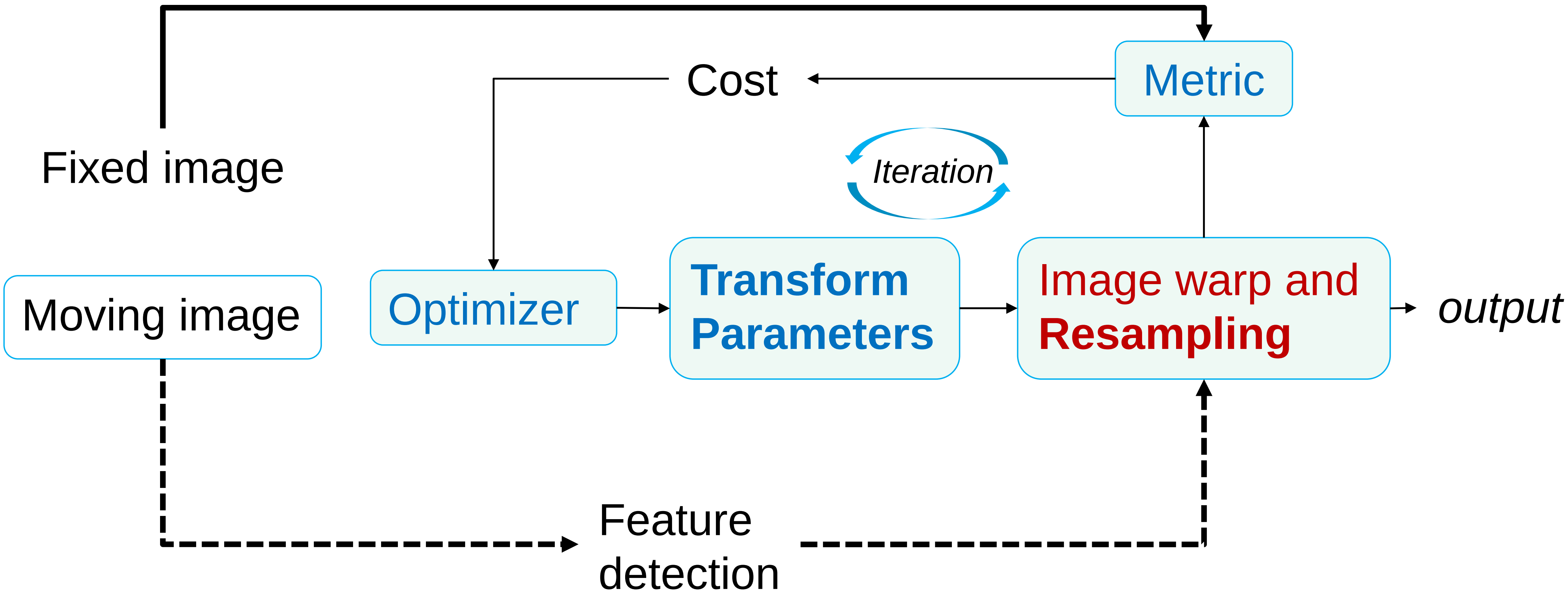




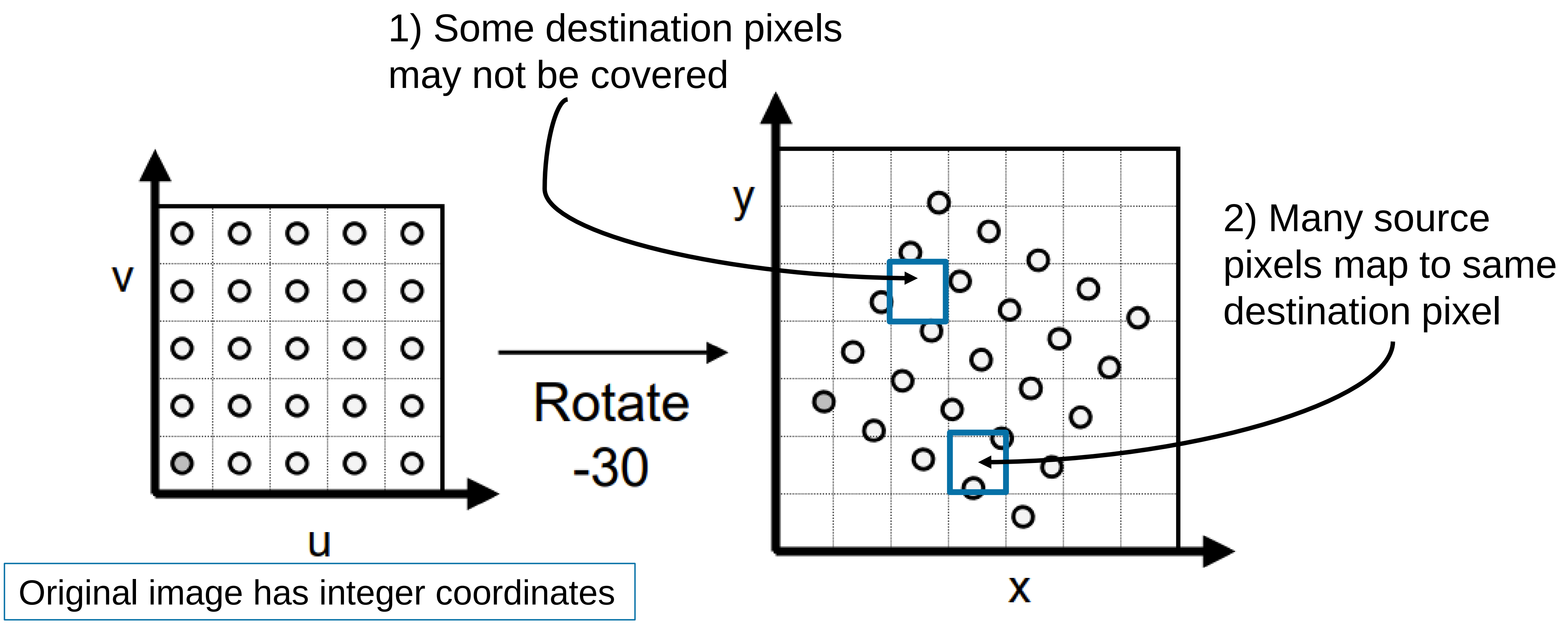




Registration Pipeline



Why interpolation

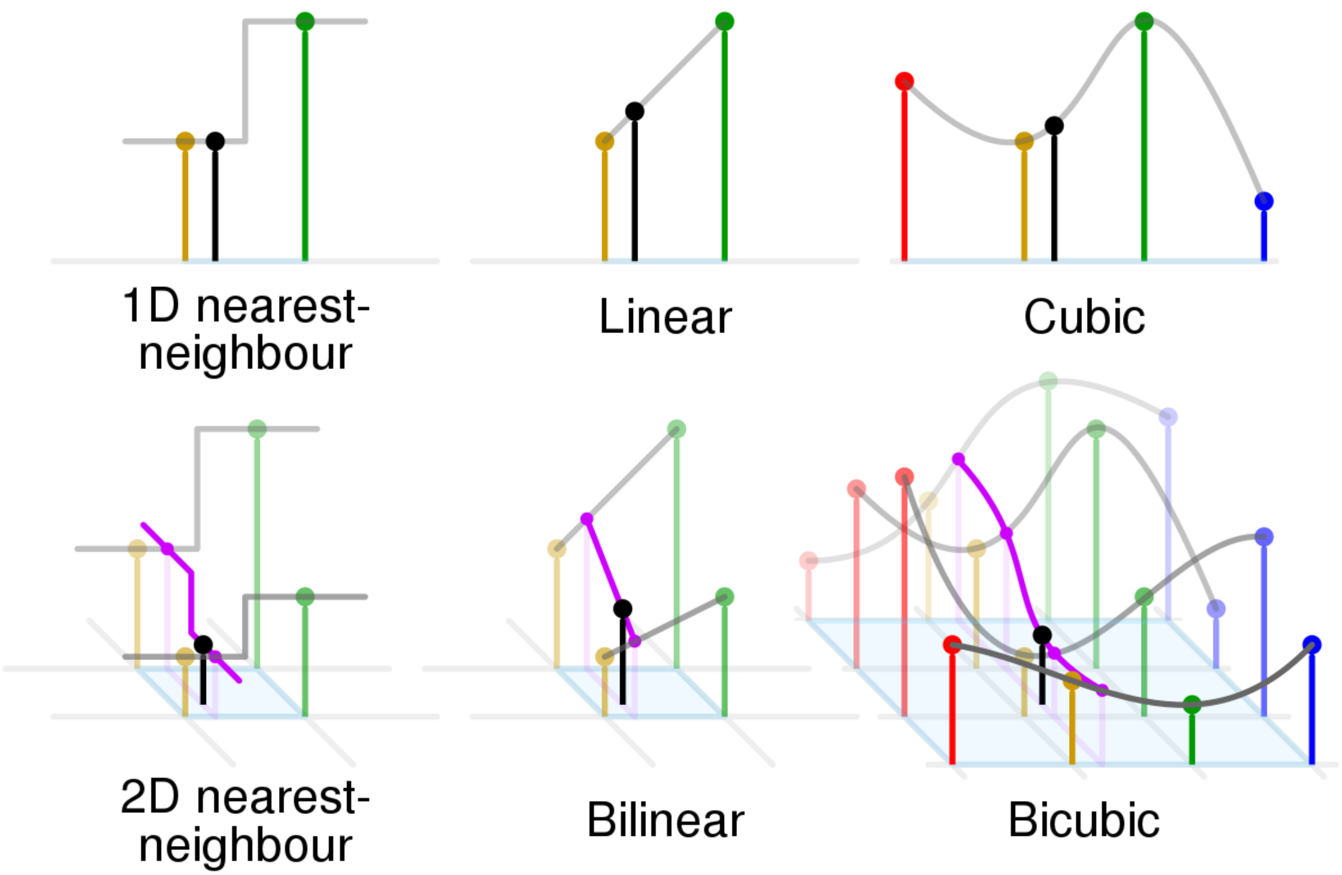


Interpolation methods (Lecture 2)

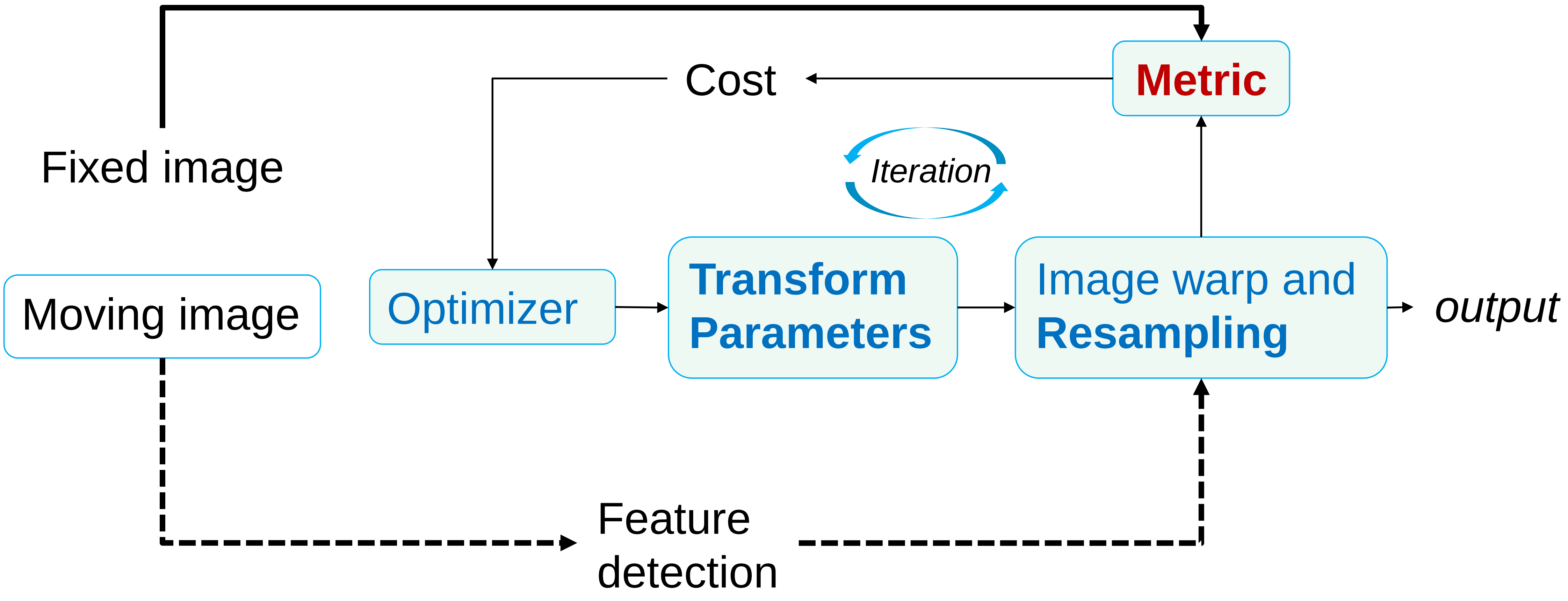
- Nearest interpolation
- Bilinear interpolation
- Bi-Cubic interpolation
- ……

2-D (16 pixels):

$$f(x,y) = \sum_{i=0}^{3}\sum_{j=0}^{3} a_{ij}x^i y^j$$



Registration Pipeline



Inter-modality & Intra-modality

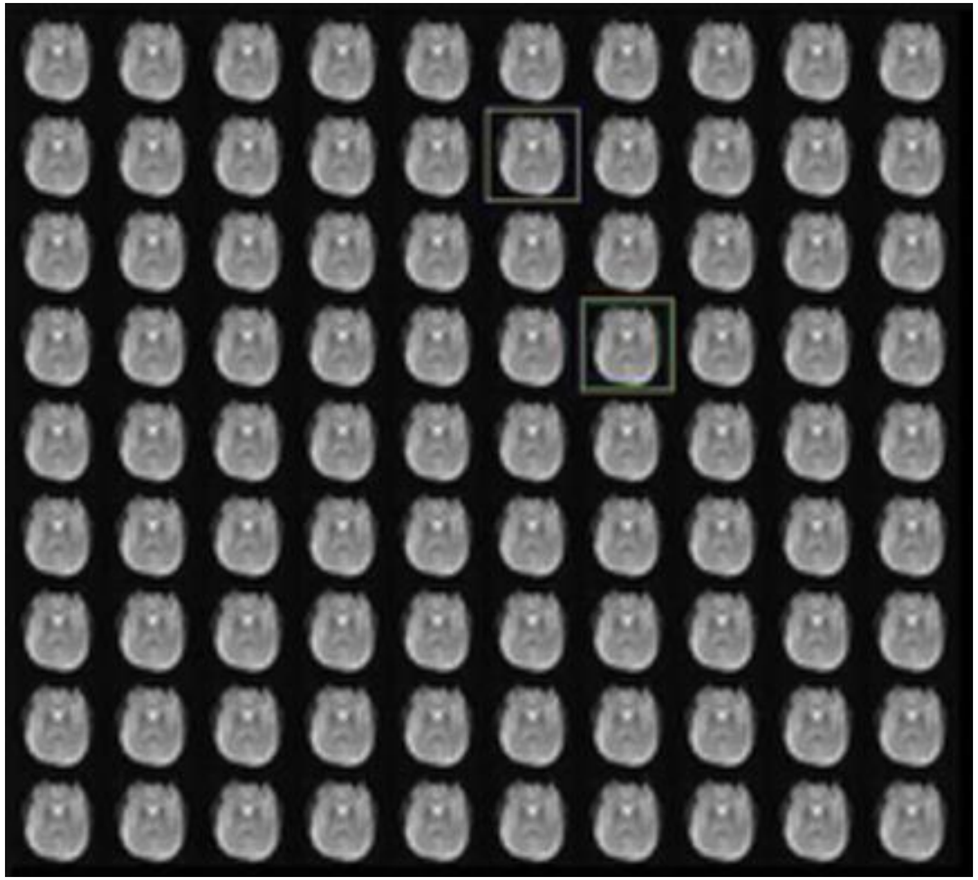
- Metrics
  - Depends on the images to be registered
- Intra-modality

- Inter-modality

Inter-modality & Intra-modality

- Alignment
  - Same modelity and same subject

BOLD fMRI



Squares of Intensity Differences (SSD)

$$\mathrm{SSD} = \frac{1}{N}\sum_{i}^{N} |A(i) - B'(i)|^2 \,, \forall i \in A \cap B'$$

Assumption

- Difference only by Gaussian noise

Work well for

- Serial MR images/ functional MR experiments

Failure

- Data diverges too much from the ideal case
- Noise
- Degradation (Motion artifacts, blurring, …)

Ratio-Image Uniformity (RIU)

Calculated over $A \cap B'$

1. Determine the number of voxels N
2. Ratio image $R(i) = \frac{B'(i)}{A(i)}$
3. Mean of R: $\mu_R = \frac{\sum_i R(i)}{N}$
4. Standard deviation of R: $\sigma_R = \sqrt{\frac{1}{N}\sum_i (R(i) - \mu_R)^2}$
5. RIU= the normalized standard deviation σ_R / μ_R

Work well for

- Useful when images differ by a gain factor
- For example, multiple PET images, serial MR registration of brain

Correlation Coefficient (CC)

$$\mathrm{CC} = \frac{\sum_i (A(i) - \bar{A})\left(B'(i) - \bar{B'}\right)}{\left\{\sum_i (A(i) - \bar{A})^2 \sum_i (B'(i) - \overline{B'})^2\right\}^{1/2}}, \forall i \in A \cap B'$$

Assumption

- A & B’ are linearly related

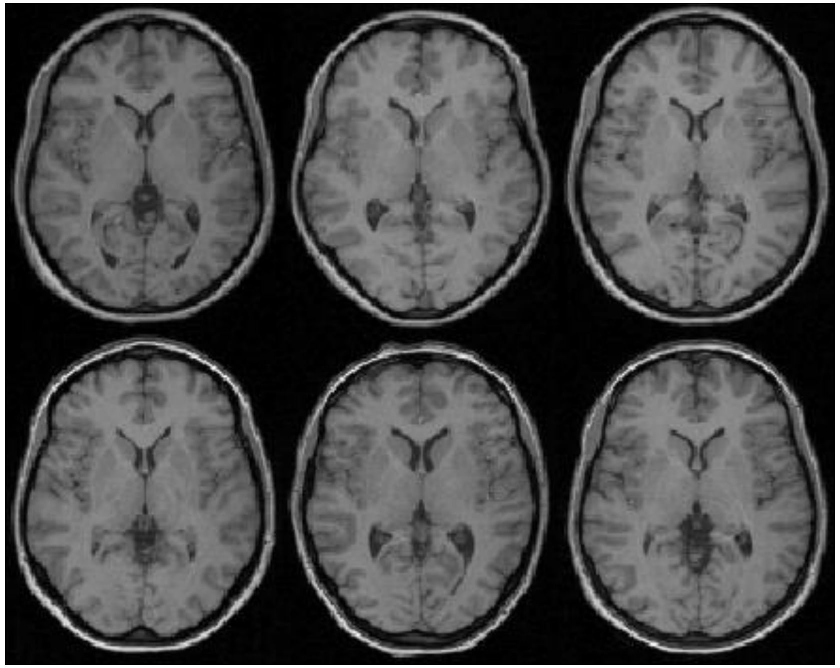
Work well

- Good for intramodal images with varying intensity scales

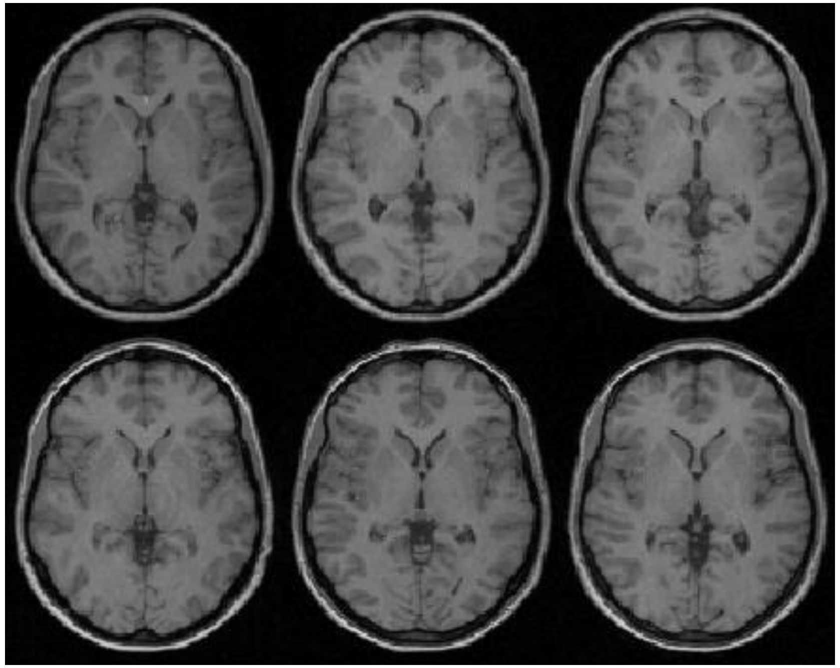
Inter-modality & Intra-modality

Intra-modal registration

- Different subjects from the same modality / camera
- Spatial normalization

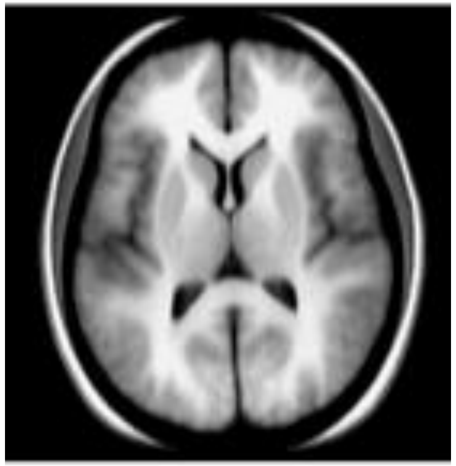


Affine registration



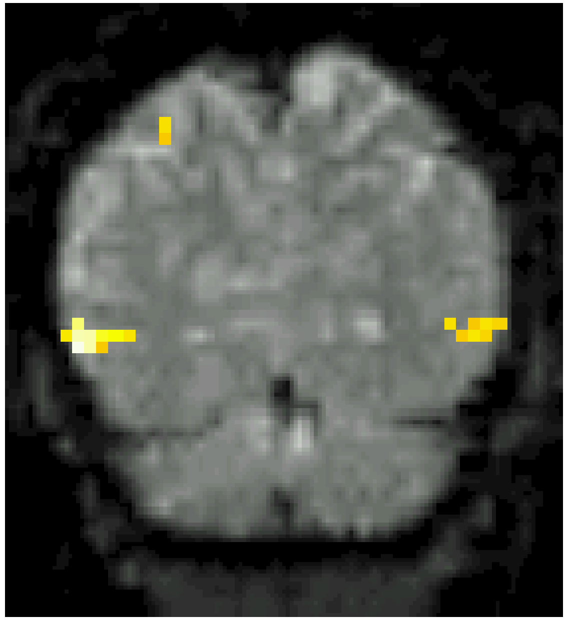
Non-linear registration

Template Images

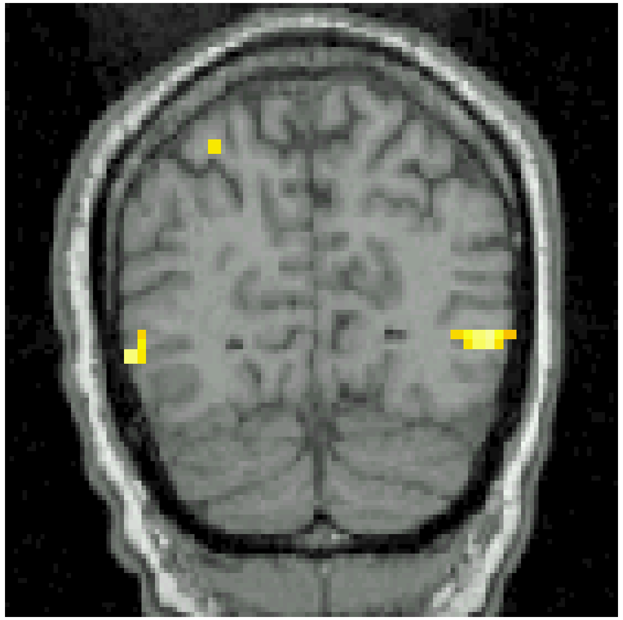


Inter-modality & Intra-modality

fMRI



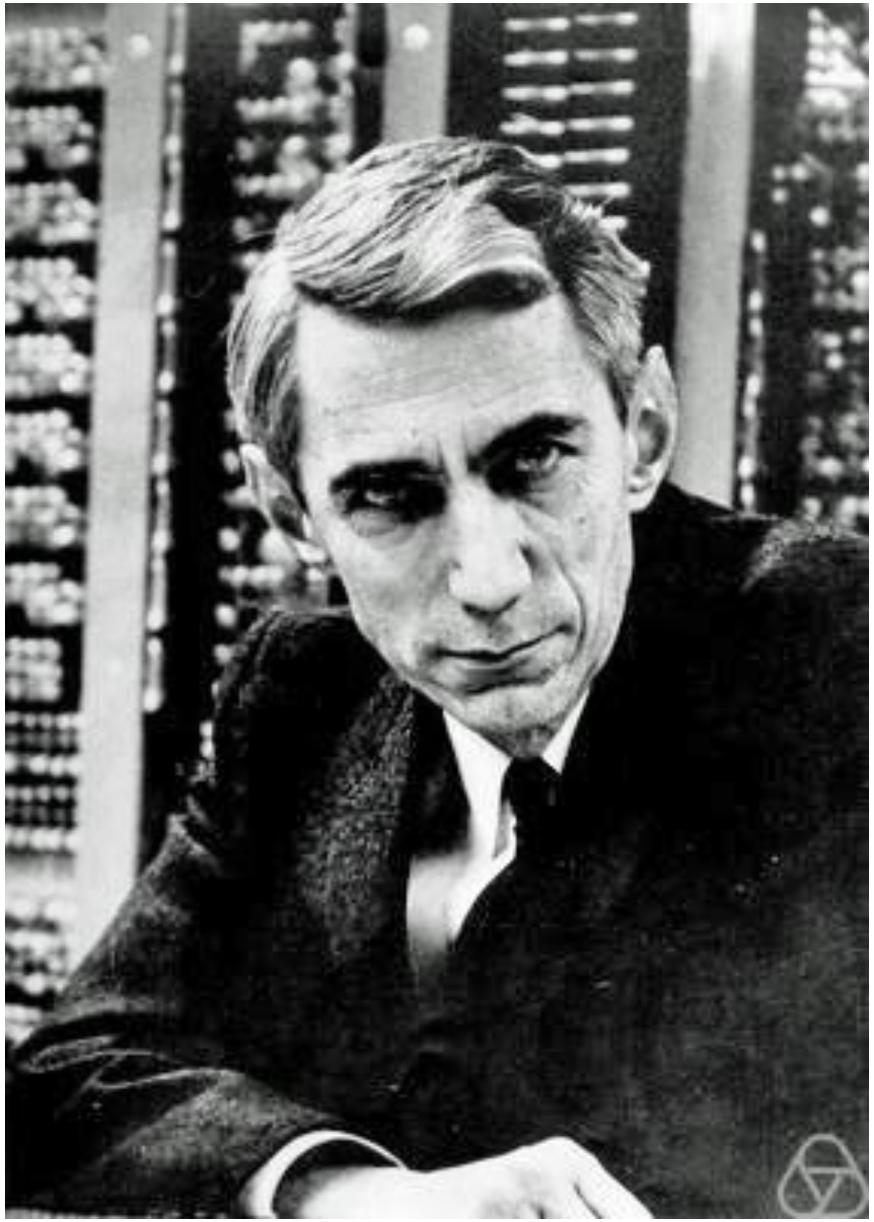
T1w



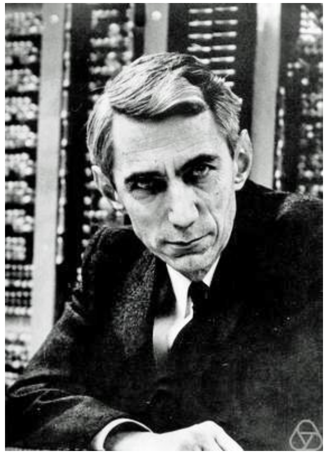
Inter-modal registration

Information

- What is information?
- Claude Shannon
  - father of information theory
- Degree of Surprise:
  - More surprise → more information
  - Surprise = Unlikely = low probability



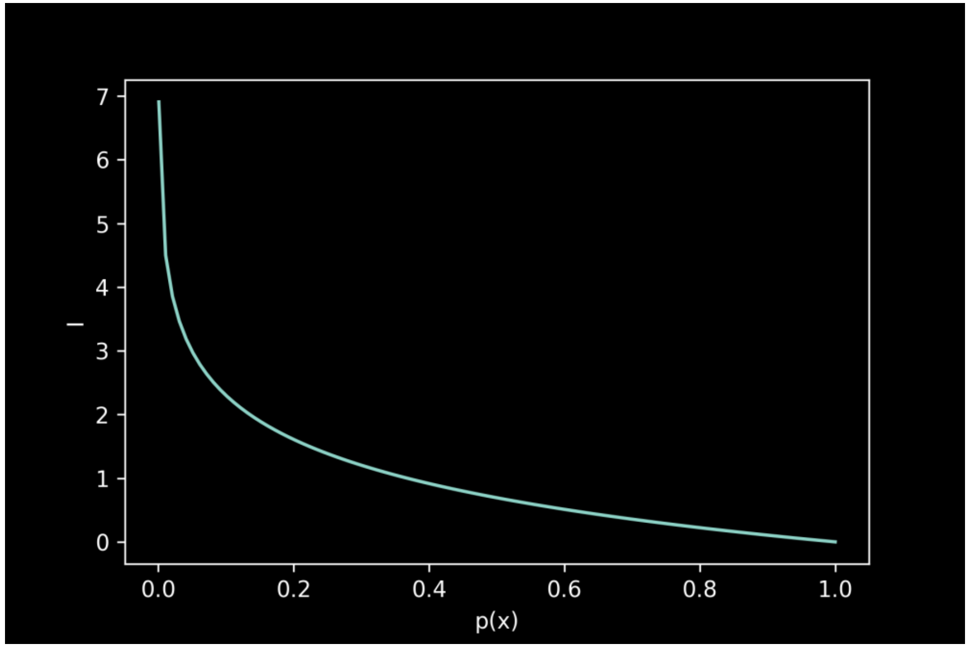
Information Theory



Claude Shannon

- Low probability = high information
  - Sun rises in the east and set in the west
  - Sun rises in the west and set in the east

Information: $I(x) \coloneqq -\log(p(x))$



- Property:
  - Continuous
  - Non-negative [0, 1]
  - Monotonic
  - Addictive

Entropy

- The expectation of information, or entropy:

$$H(X) := E[I(x)] = \sum p(x)I(x) = -\sum p(x)\log p(x)$$

- Joint entropy:

$$H(X,Y) := -\sum p(x,y)\log p(x,y)$$

Entropy

- Conditional entropy

$$H(Y|X) = -\sum p(x)H(Y|X=x)$$

$$= -\sum_{x} p(x) \sum_{y} p(y|x) \log p(y|x)$$

$$= -\sum_{x,y} p(x,y) \log p(y|x)$$

Mutual Information

- Relative entropy: the distance betweem distribution q and p

$$D(p||q) = -\sum p(x)\log q(x) - \left[-\sum p(x)\log p(x)\right] = \sum p(x)\log\frac{p(x)}{q(x)}$$

- Mutual information = Relative entropy between the joint distribution $p(x,y)$ and the product of the distributions $p(x)p(y)$

$$I(X;Y) = H\big(p(X)p(Y)\big) - H(p(X,Y)) \qquad p \coloneqq p(X,Y), q \coloneqq p(x)p(y)$$

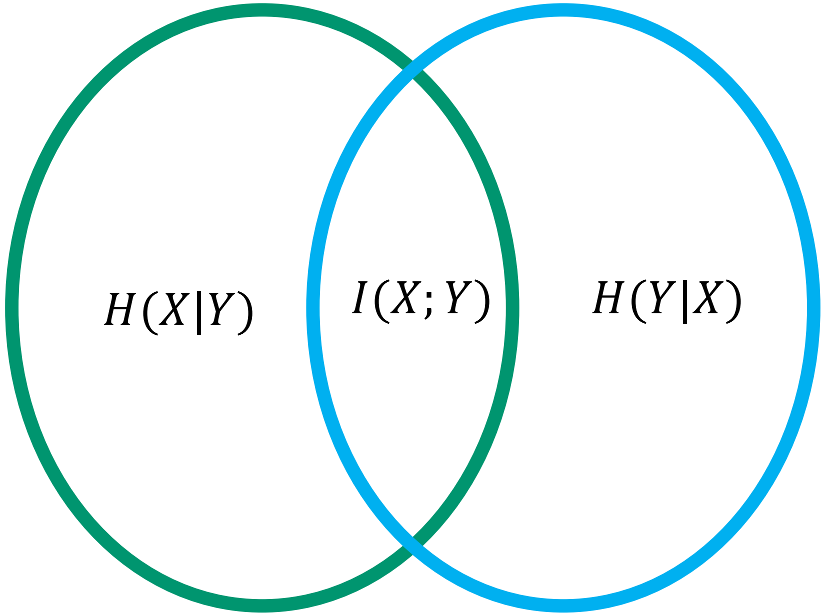
$$= \sum p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$$

- Amount of information that X contained in Y
- If Y is given, the information reduction in X

Mutual Information

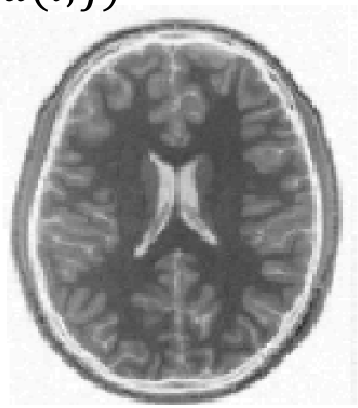
$$I(X;Y) = \sum p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$$
$$= \sum p(x,y)\log\frac{p(x|y)}{p(x)}$$
$$= -\sum p(x,y)\log p(x)$$
$$= H(X) - \underbrace{H(X|Y)}_{\text{Conditional entropy}}$$

$$I(X;Y) = H(X) + H(Y) - \underbrace{H(X,Y)}_{\text{Joint entropy}}$$

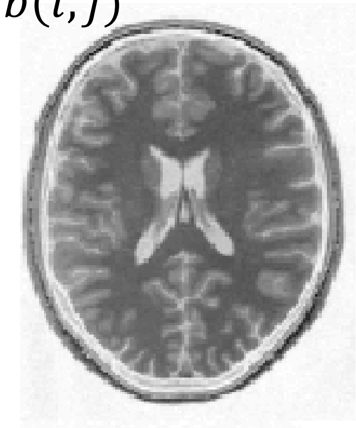


Joint Histogram

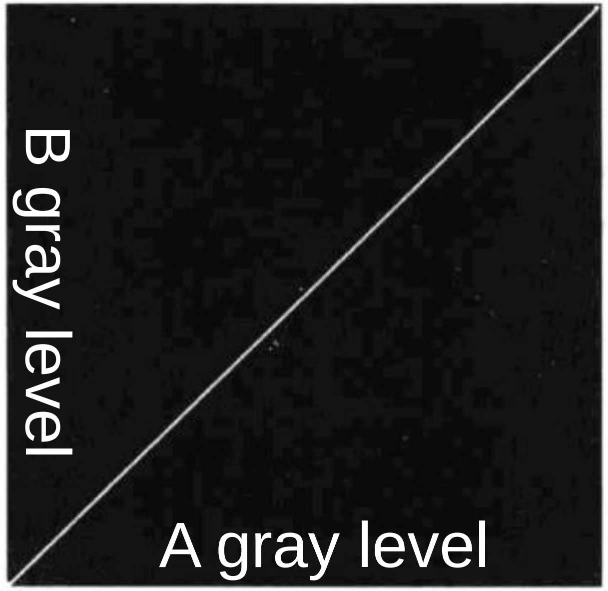
$a(i,j)$



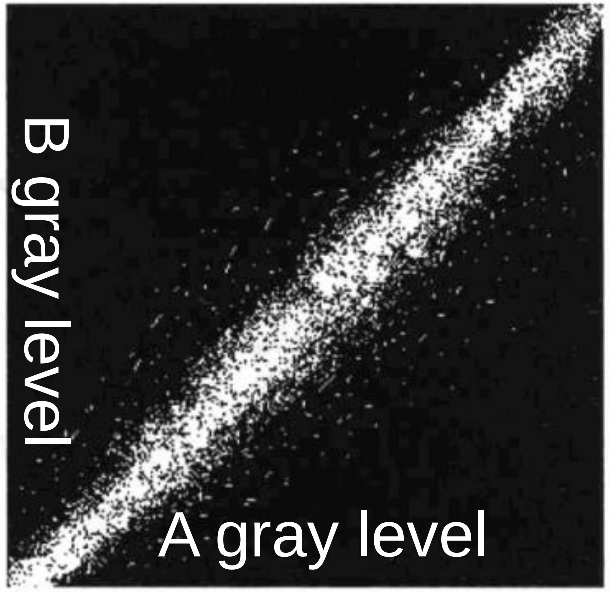
$b(i,j)$



$$h_{ab}[a(i,j),b(i,j)] = n_{[a(i,j),b(i,j)]}$$



The identity

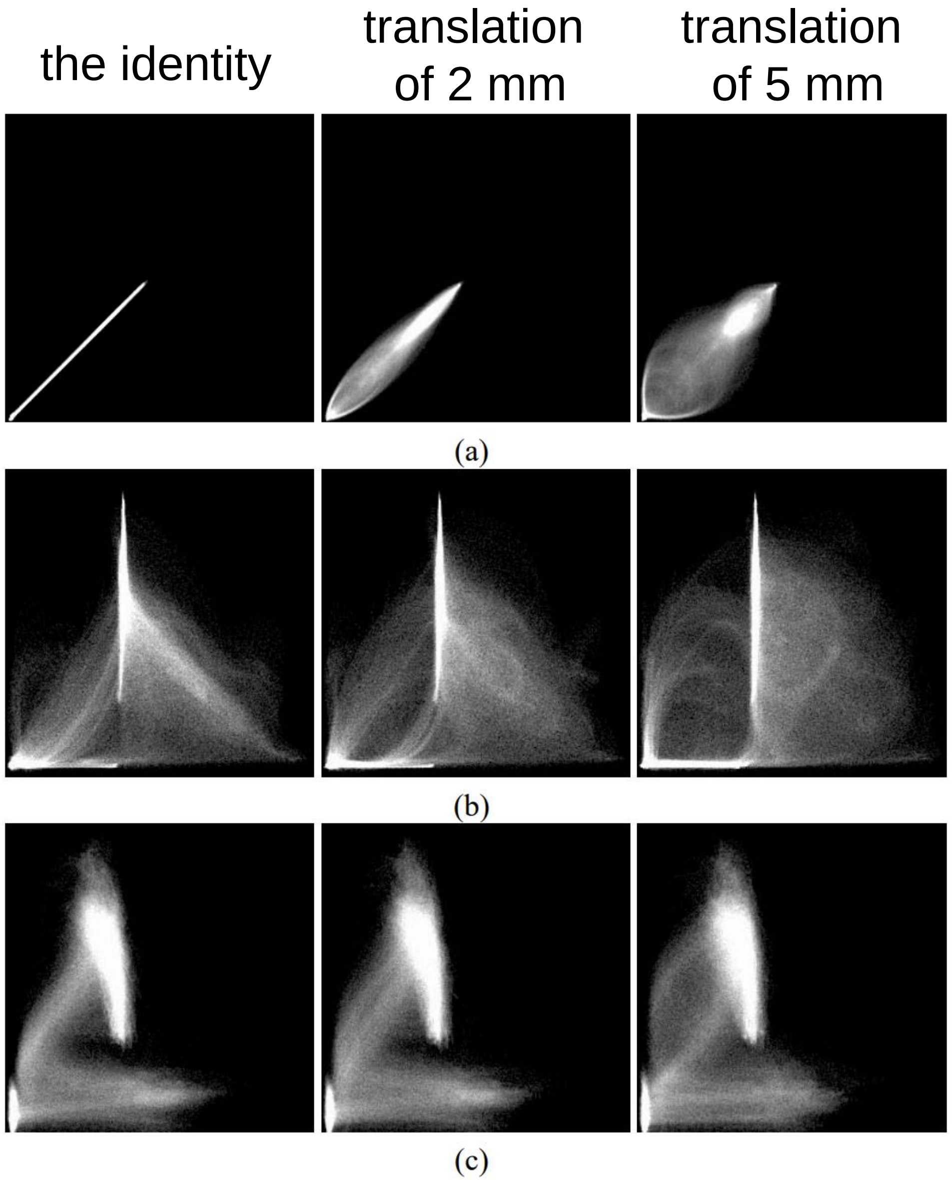


Lower similarity

Normalized Joint Histogram (Joint probability distribution function, **PDF**)

$$p_{ab}[a(i,j),b(i,j)] = \frac{h_{ab}[a(i,j),b(i,j)]}{\sum_{i,j} h_{ab}\,[a(i,j),b(i,j)]}$$

Example joint PDF



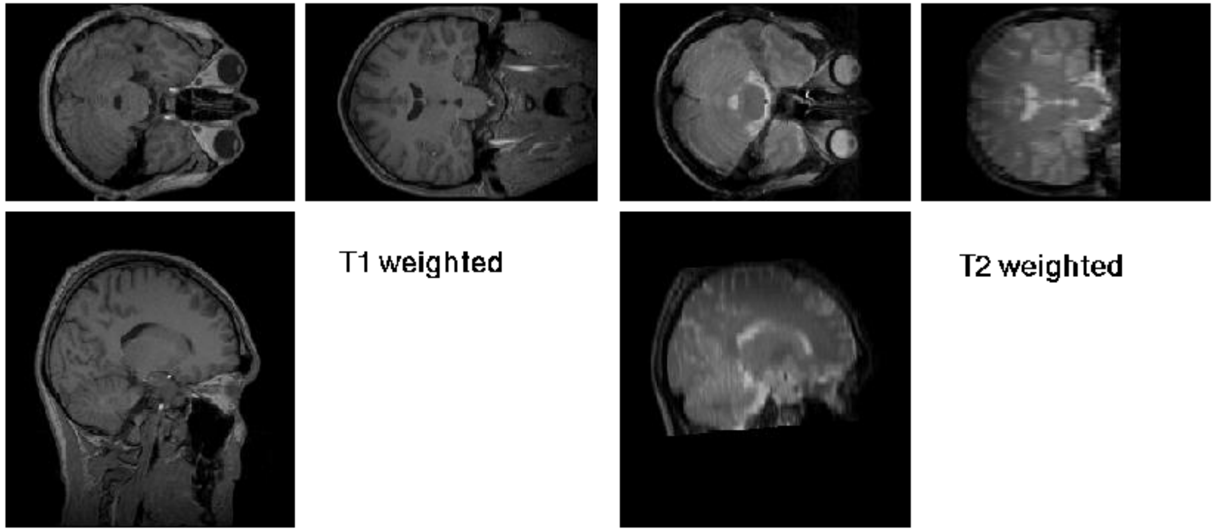
Identical MR image

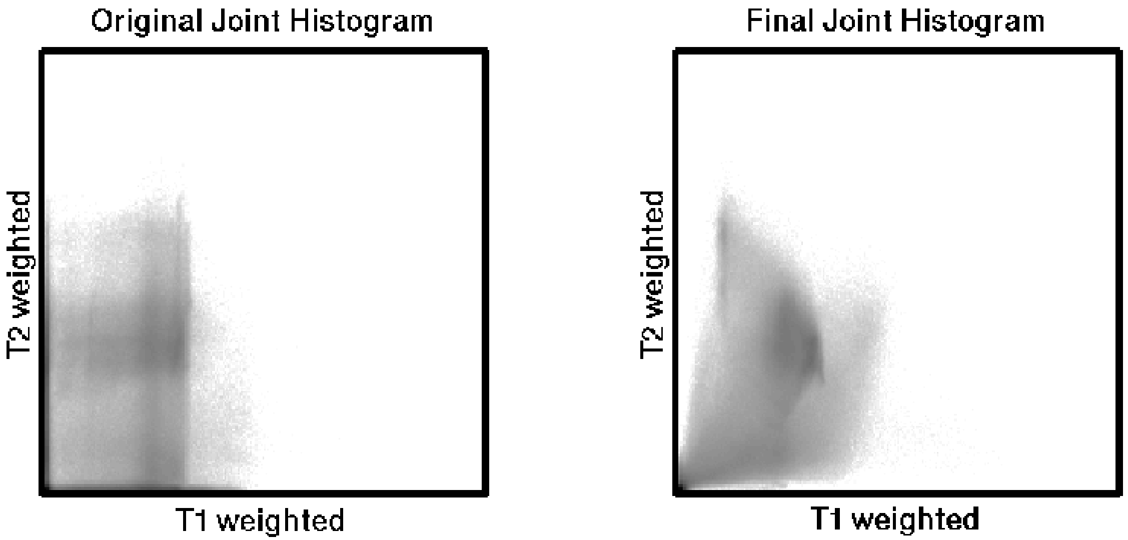
CT and MRI

MR and PET

Mutual Information (MI) as a measure

- Correct registration results in a **decrease in joint entropy, and increase of MI**





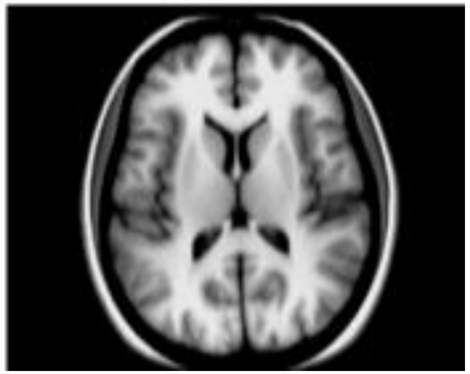
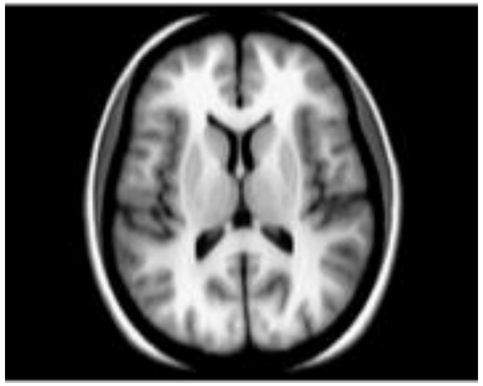
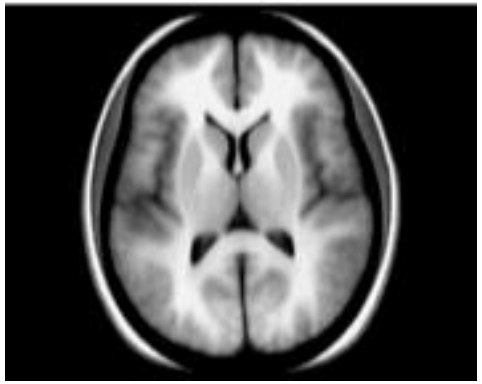
- Good for both Intra & Inter-modality registration

Summary

Cost Function	Modality Assumption	Robustness	Computational Cost	Best Use Case
SSD / MSE	Intramodal	Low (noise-sensitive)	Low	Same modality, no intensity changes
RIU	Intramodal (scaled)	Moderate	Moderate	Same modality, multiplicative differences
Cross-Correlation	Intramodal / Linear	High (linear changes)	Medium	Same modality, brightness/contrast changes
Mutual Information	Multimodal	Very High	High	Different modalities

Example: Spatial normalization

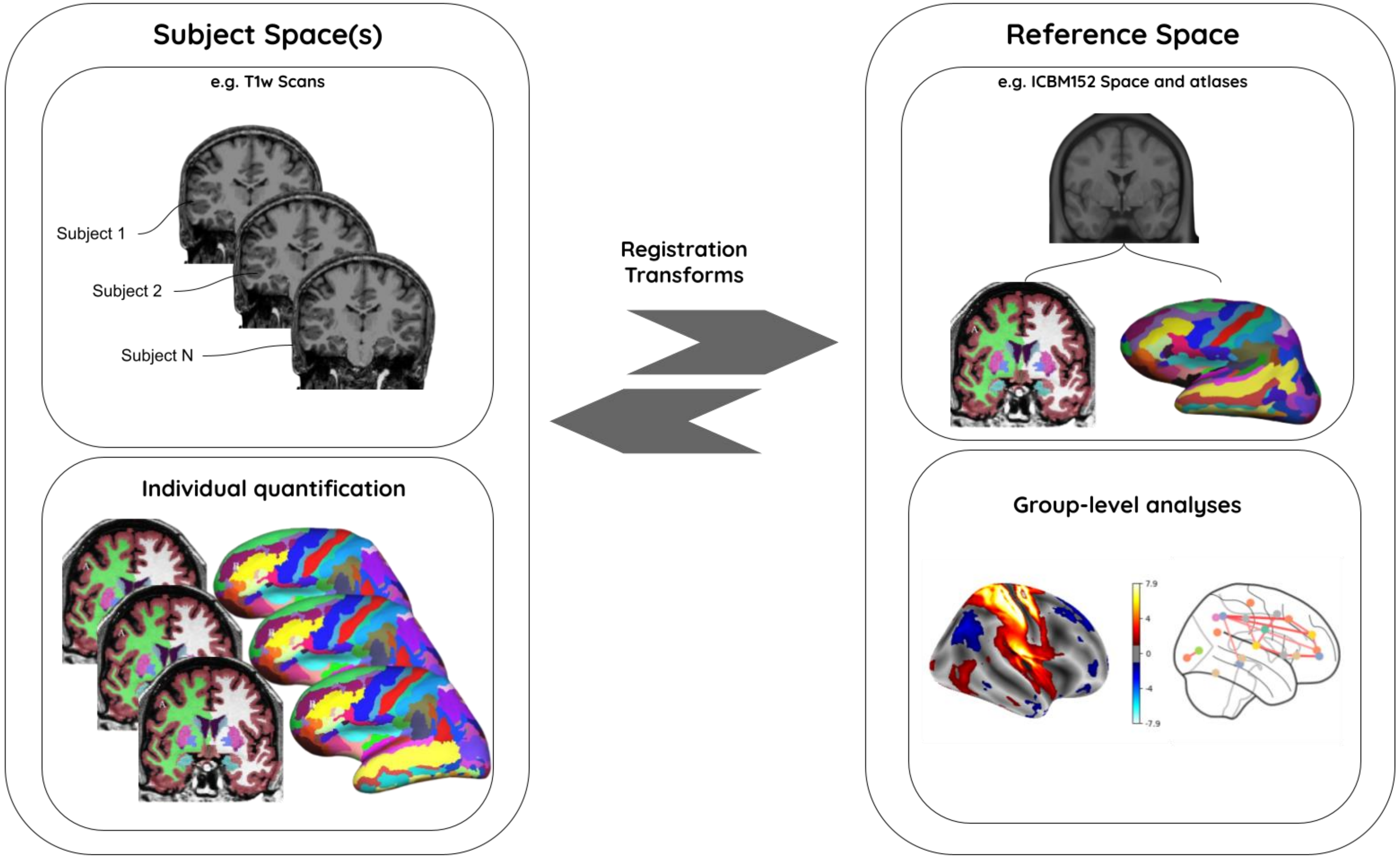
Spatial normalization of 20 subjects

Mean Square Error	Cross correlation	Mutual Information
		
		reduced acutance

Avants BB, Tustison NJ, Song G, Cook PA, Klein A, Gee JC. A reproducible evaluation of ANTs similarity metric performance in brain image registration. Neuroimage. 2011 Feb 1;54(3):2033-44.

Spatial Normalization

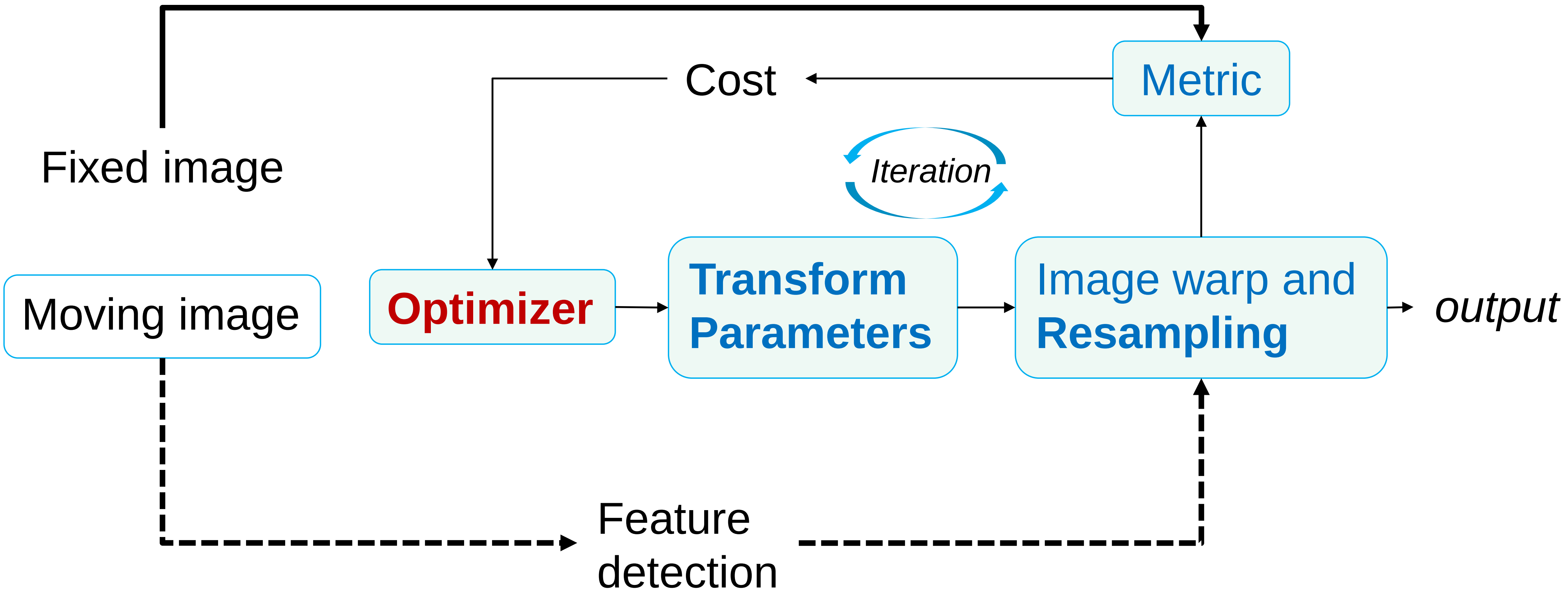
- Application: subject to reference



Contents

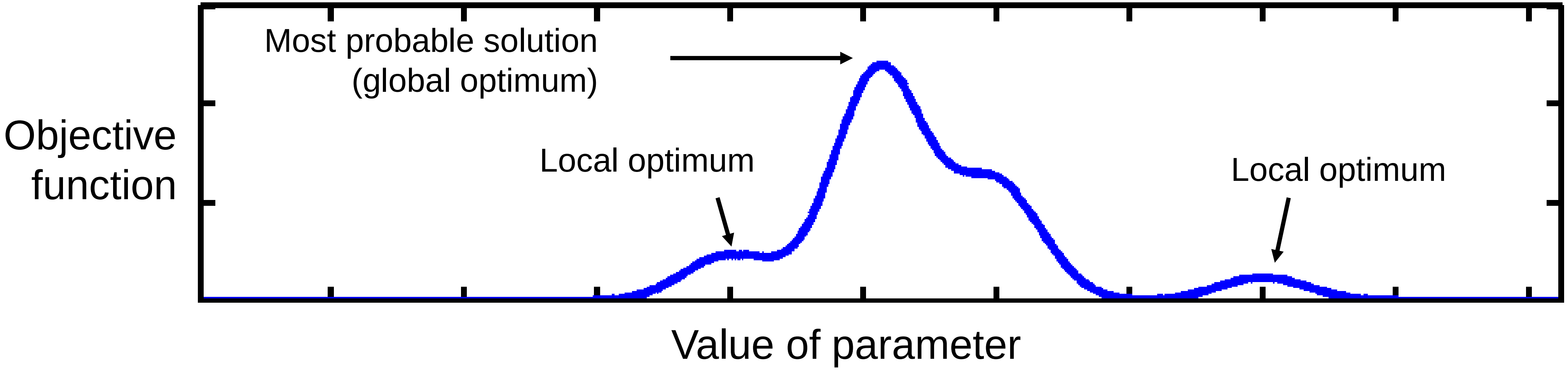
- Linear registration
  - Tramsformation
  - Similarity metric
  - Optimization
- Non-linear registration
  - Diffeomorphic transformation
  - Jacobian matrix
- Toolbox and pipeline

Registration Pipeline



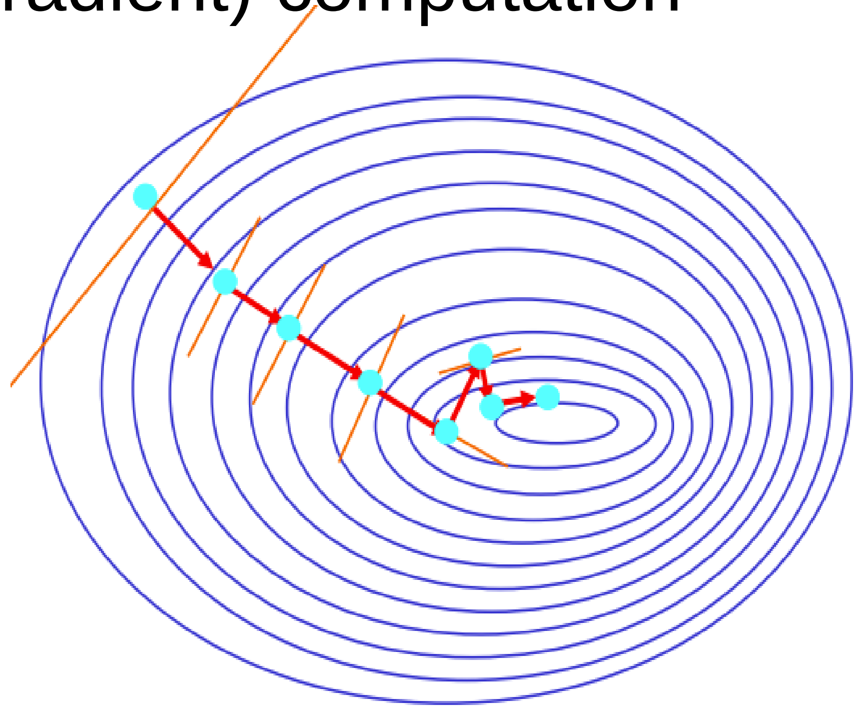
Optimization Approach

- Optimization: finding some “best” parameters according to an “cost function”
- The “cost function” is often related to a probability based on some model



Optimization Approach

- Closed form solution does not exist
- Cost functions must be minimized iteratively (brute force or more intelligently)
- Optimization require derivative (or gradient) computation
  - *One-dimensional search*
  - *Gradient descent*
  - *Variable metric*



Optimization Approach

Similarity Index/Object function

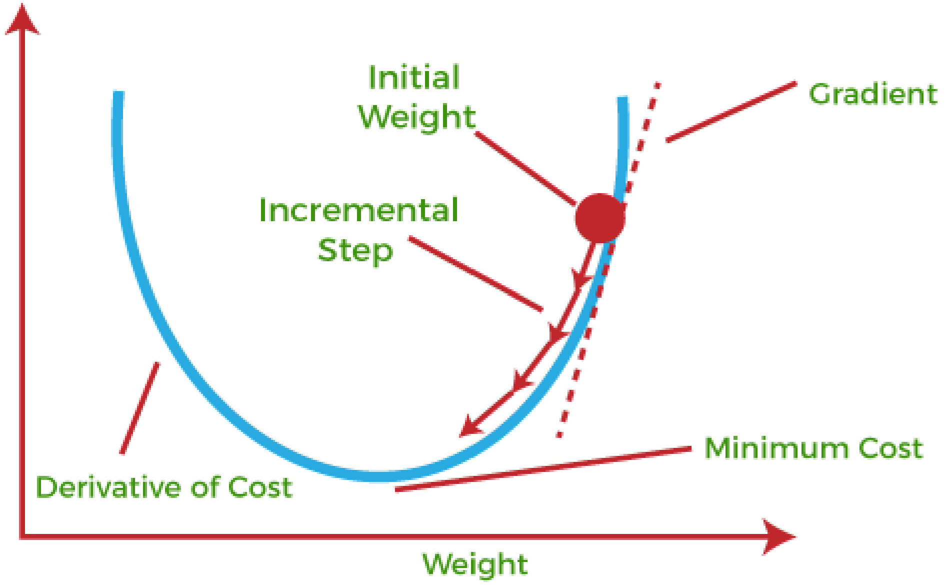
- Intra-modal
  - Mean squared difference (minimise)
  - Normalised cross correlation (maximise)
  - Entropy of difference (minimise)

- Inter-modal (or intra-modal)
  - Mutual information (maximise)
  - Normalised mutual information (maximise)

Example Approach : Gradient Descent

- Consider any registration similarity index C as a function of two images and M transformation parameters

$$\nabla C(A, B; \boldsymbol{p}) = [\frac{\partial C}{\partial p_1}, \dots, \frac{\partial C}{\partial p_M}]$$



- A gradient descent update is given by

$$p_{t+1} = p_t - \eta \nabla C(A, B; \boldsymbol{p})$$

step size

- Check if new parameter improves similarity index, if not then reduce step size, and repeat

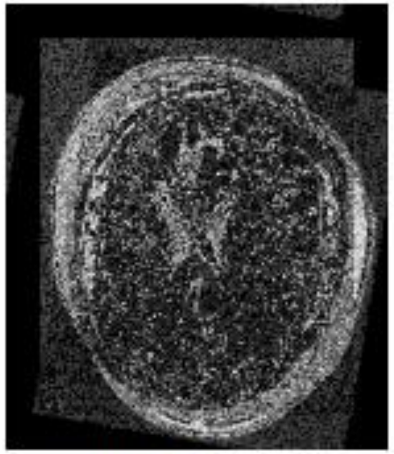
Example Approach : Gradient Descent

- Numerical approximation

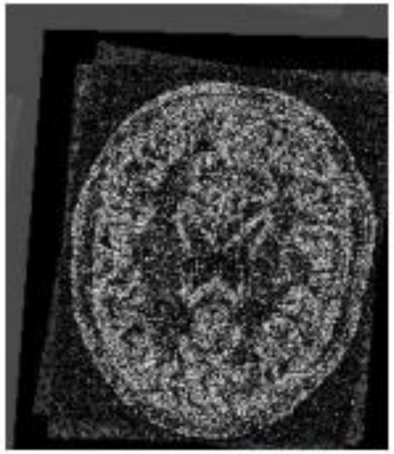
$$\frac{\partial C}{\partial p_i} \approx \frac{C(p_1, \dots, p_i + \Delta p_i, \dots, p_M) - C(\boldsymbol{p})}{\Delta p_i A}$$

  - Choice of delta will affect results
  - Can always be computed
  - Better alternative when possible: directly computing derivative from the cost function

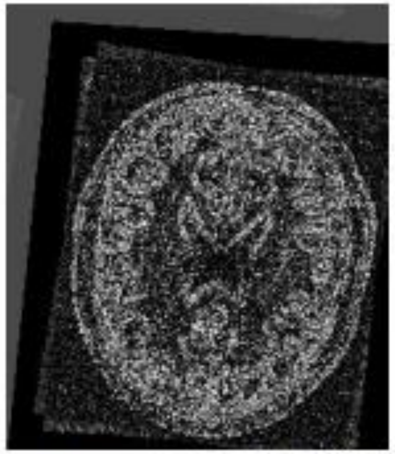
Registration result



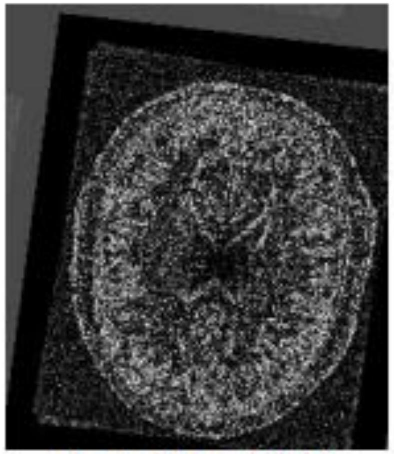
Initial errors



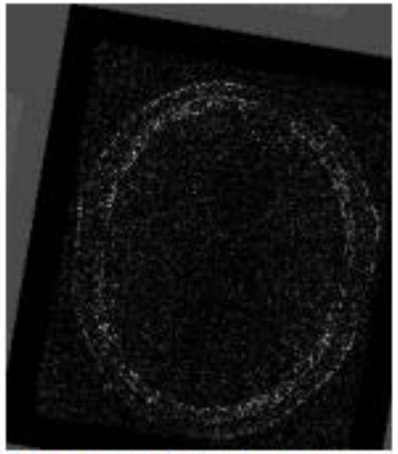
Iteration 100



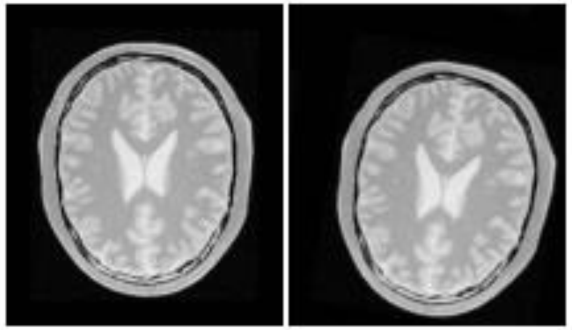
Iteration 200



Iteration 300



Final: 498 iterations

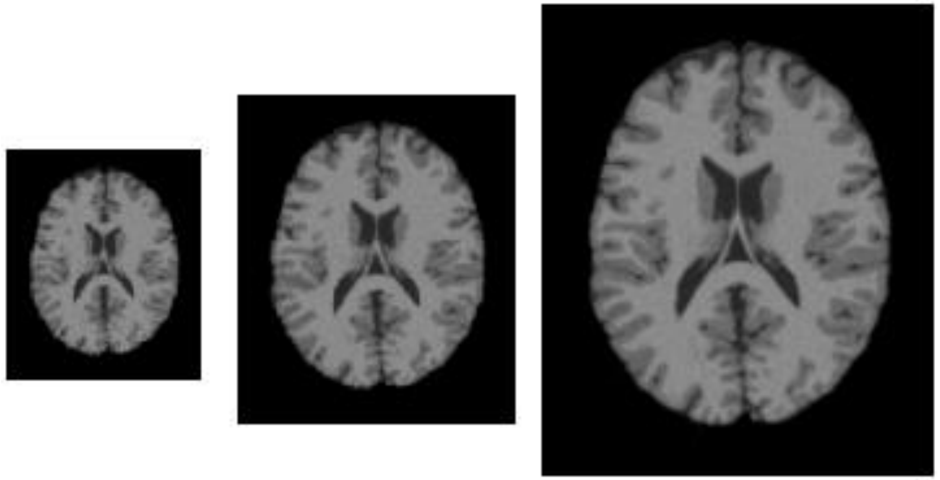


Practice Issues

$$A(m,n) - sB(T(m,n;\boldsymbol{p}))$$

- For SSD registration, an intensity **scaling** parameter is often included
- Sensitive to initialization, "multi-start" approaches
- In order to avoid local minima, **a multi-resolution scheme** is typically employed

- First run at a coarse resolution (e.g. resampled to 6x6x6 mm)
- Second stage resolution (e.g. resampled to 3x3x3 mm)
- Repeated once more with 1.5x1.5x1.5mm

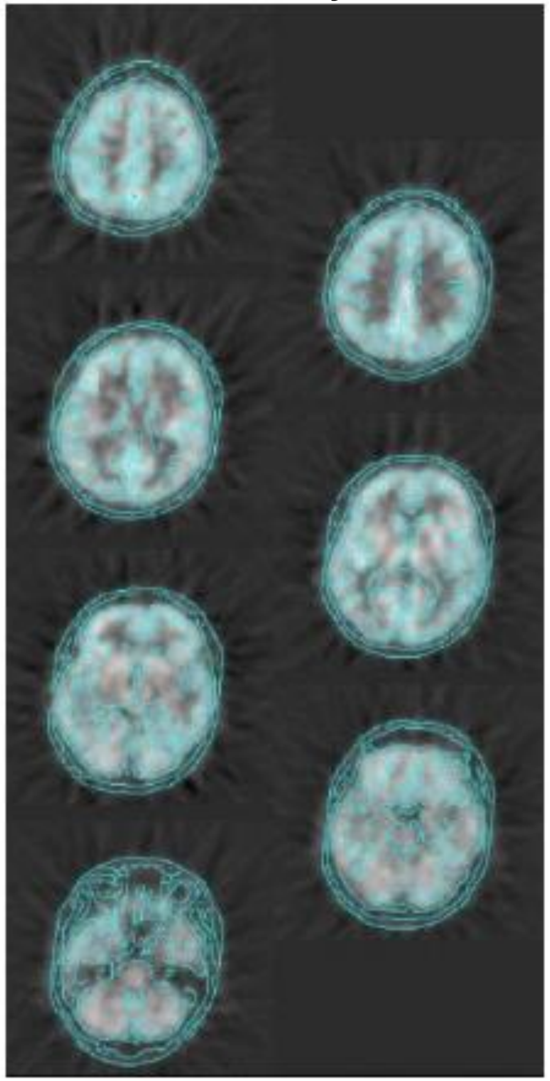


Evaluation of registration algorithms

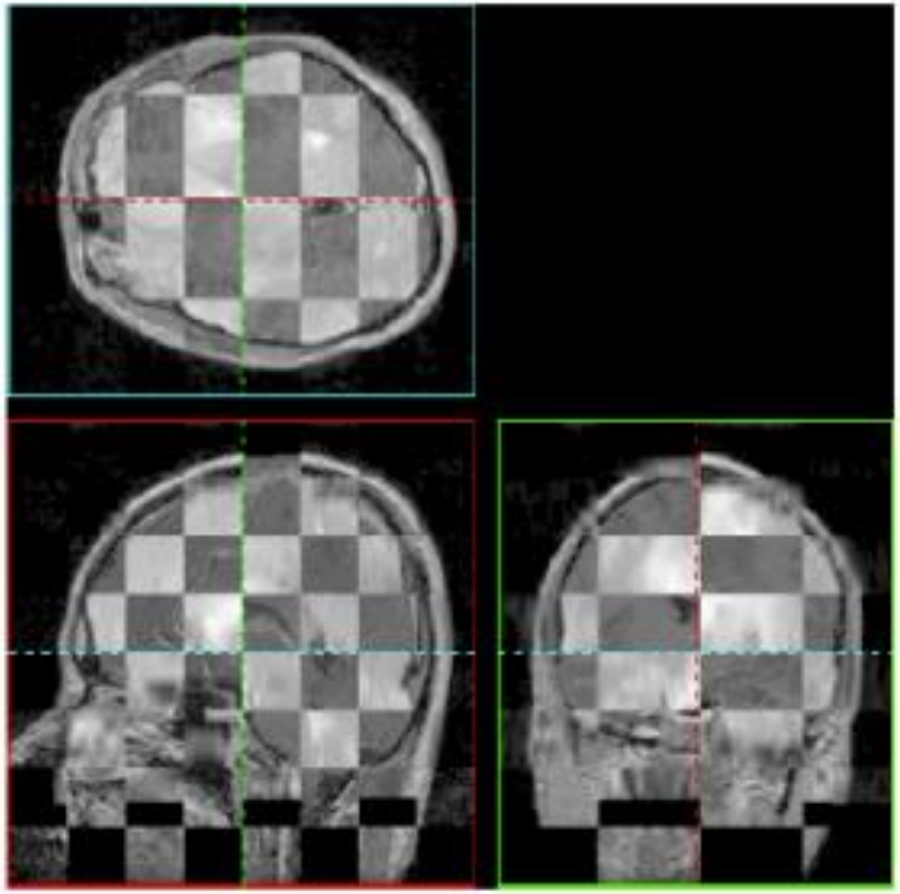
- A gold standard is available
  - Derived from simulated data or fiducial markers
- User selected ***landmarks*** or ***manually defined structures*** may be used to validate the registration result
- Visual assessment (overlays, checkerboard patterns, average images)

Visual assessment

Overlay



Check board



Registration of SPGR and T2 images

Contents

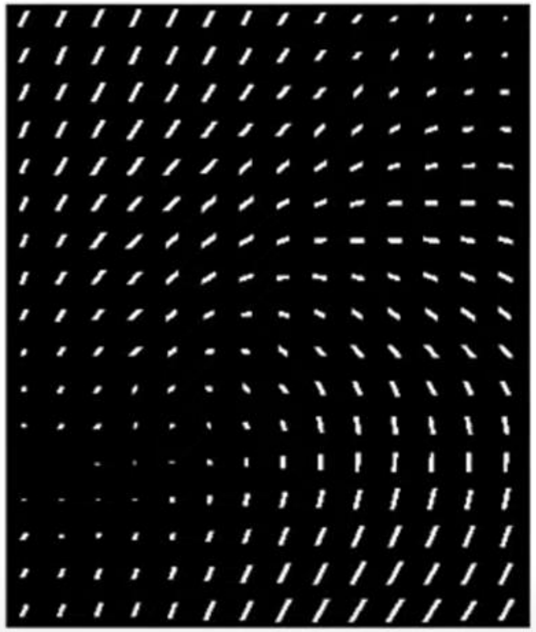
- Linear registration
  - Previously: interpolation & spatial transform
  - Similarity metric
  - Optimization
- Non-linear registration
  - Diffeomorphic transformation
  - Jacobian matrix
- Toolbox and pipeline

Nonlinear Registration

- One approach is to use a parametric representation of the deformation field and optimize similarity functions w.r.t. those parameters
- Another approach is to break down transformation into smaller affine/rigid transformations and constructing a smooth deformation from those transformations
- Issues such as initialization and local minima become even more prominent

Nonlinear Registration Methods

A nonlinear transformation can be represented by a **deformation field**

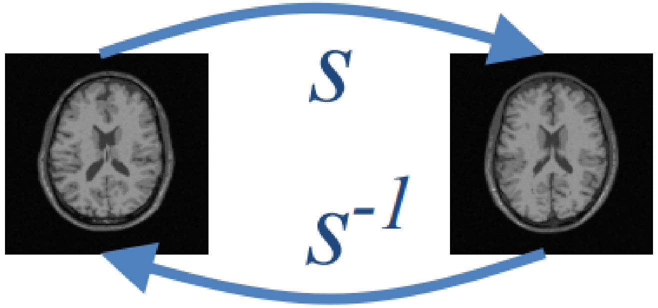


Deformation Field $u\colon \Omega_I \to \mathbb{R}^3$

- Diffeomorphic
  - Demons
  - LDDMM
  - HARMMER
- Free-form Deformation
- ……

Diffeomorphic Deformation

- In mathematics, a **diffeomorphism** is an isomorphism of smooth **manifolds**.
- An invertible function that maps one differentiable manifold to another such that both the function and its inverse are differentiable.
- **Diffeomorphic mapping** is any solution that registers or builds correspondences between images by ensuring the solutions (transformation) are diffeomorphic
  - Key element: deformations = composition of a number of small deformations
  - One to one, invertible, smooth
  - Preserves topology
  - Essential for computational anatomy



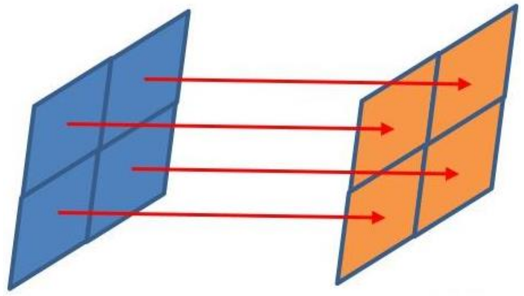
Diffeomorphic Deformation

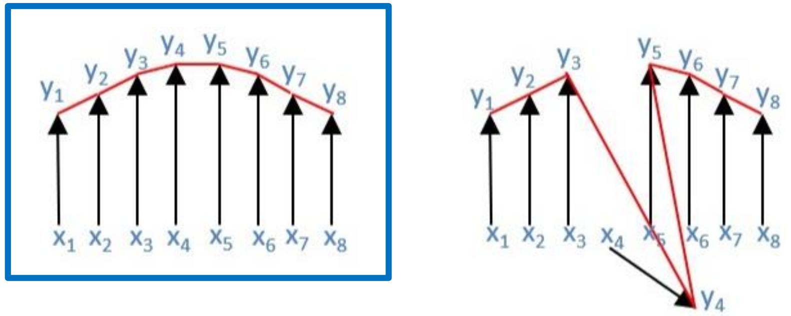
Diffeomorphic mapping

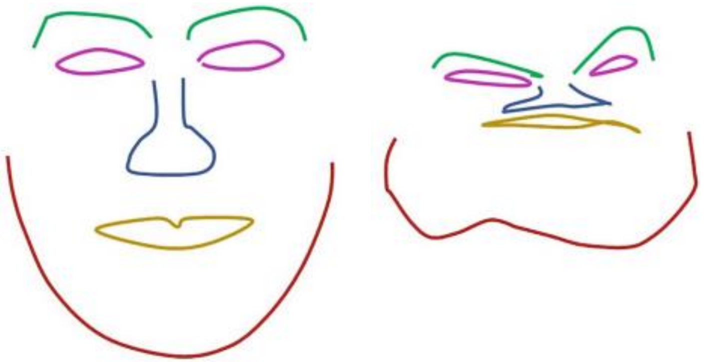
Homeomorphism

Differentiable

Topology



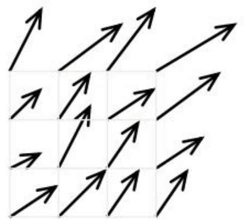




One-to-one mapping

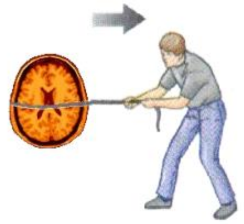
Smooth & differential mapping

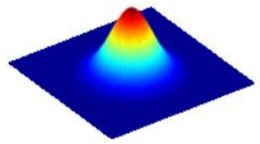
Classical Demons Algorithm



- Using more convenient notation
  - F, M: fixed and moving images
- Demons iterations
  - Initial displacement field s (e.g., identity deformation)
  - Start with all pixels in M whose spatial gradient is not zero: D_M
  - Compute the elementary displacement $u(P)$ for all P in D_M
  - Compute image forces u to push $M \circ s$ towards F

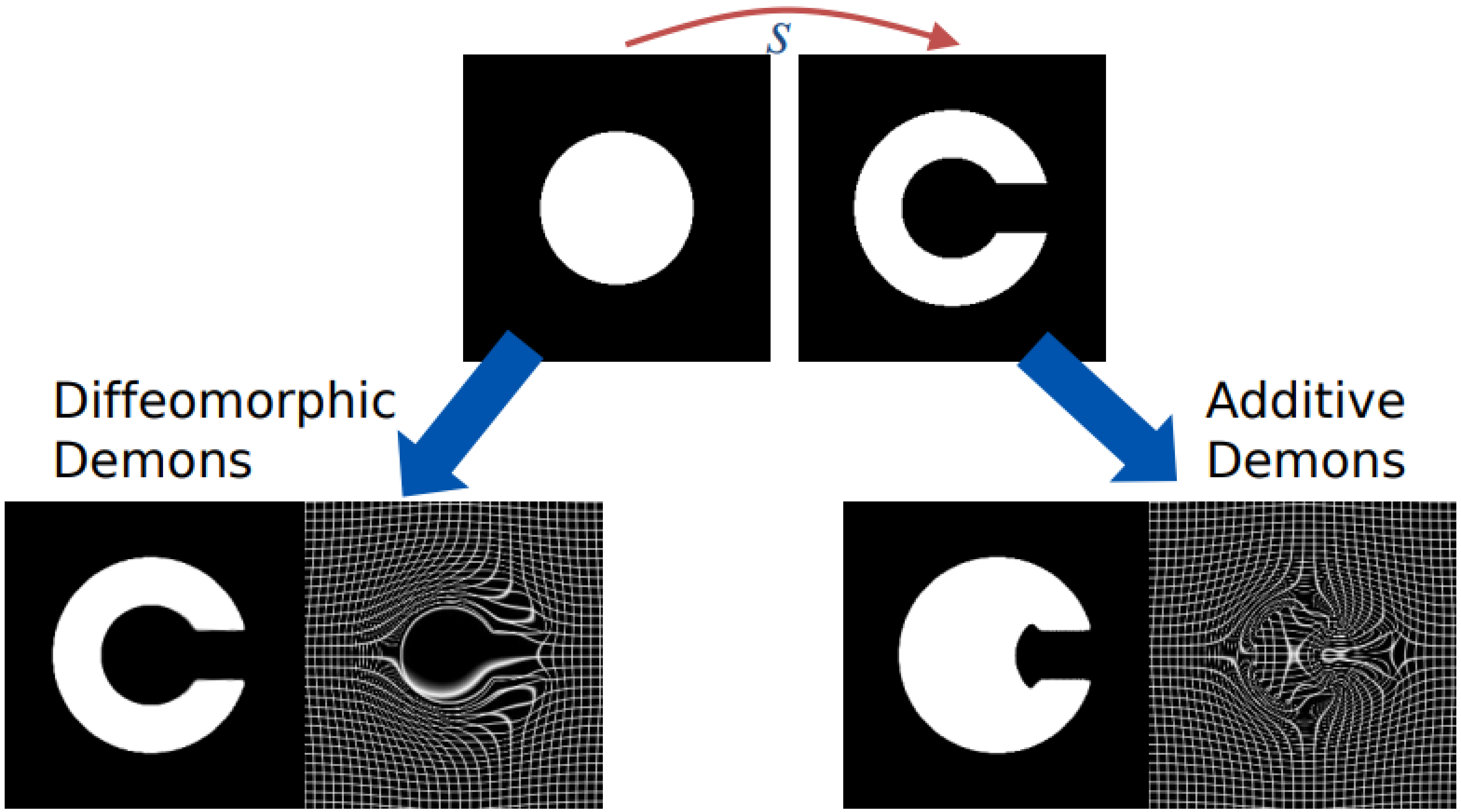
    i.e. make $M \circ (s + u)$ more similar to F
  - $s \leftarrow$ Gaussian smoothing on $s + u$





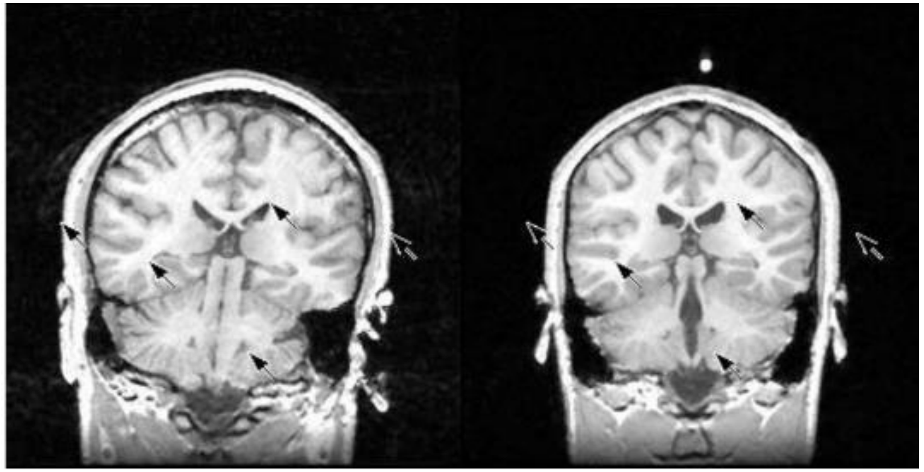
Note: intuition behind the image force: Demons push according to image gradient if pixel value is lower than target value. Opposite of the image gradient if pixel value is higher than target value.

Example

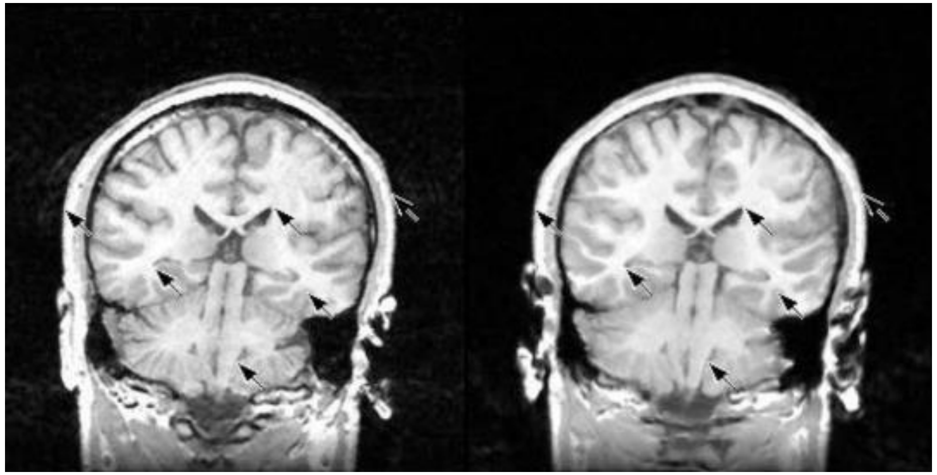


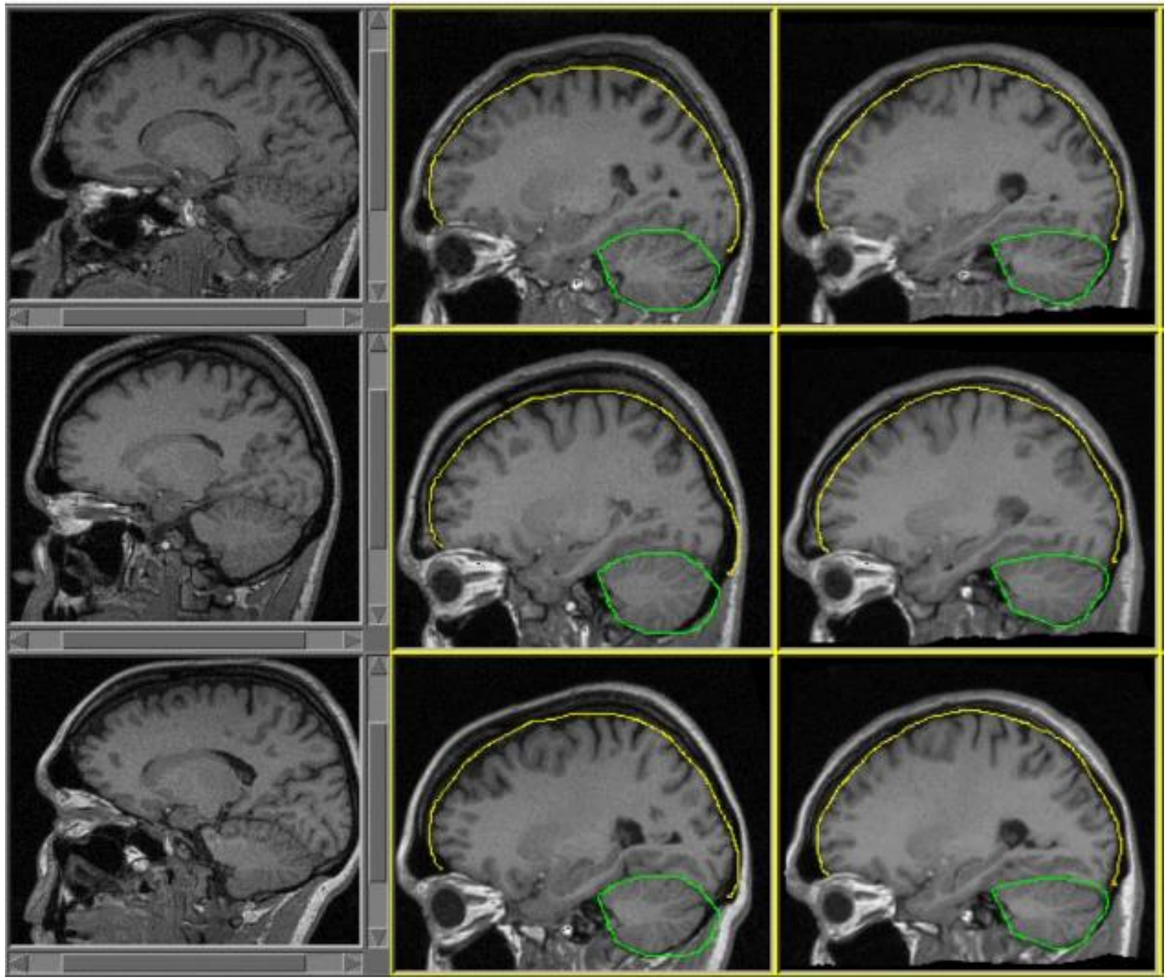
Example

Pairwise registration problem

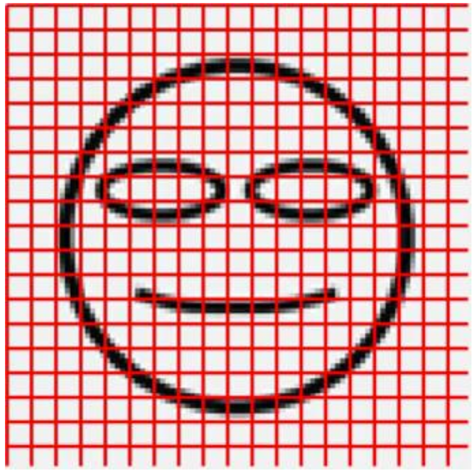


Demons result

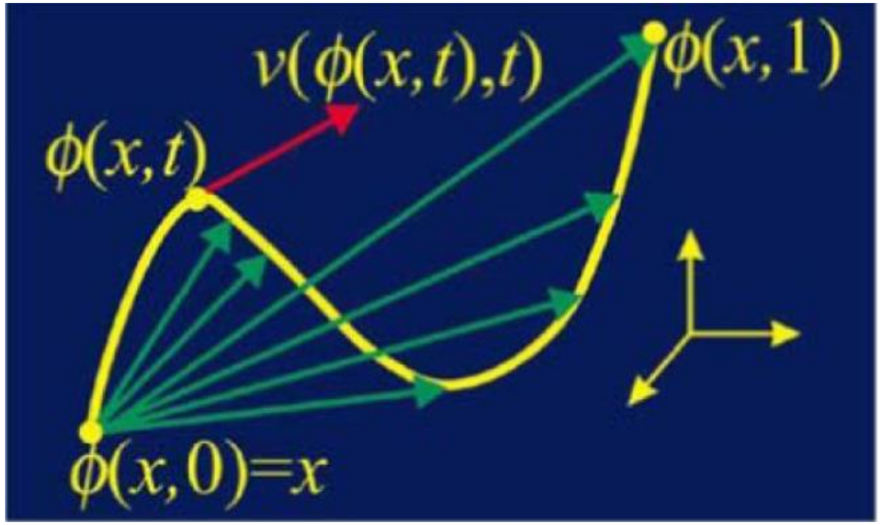


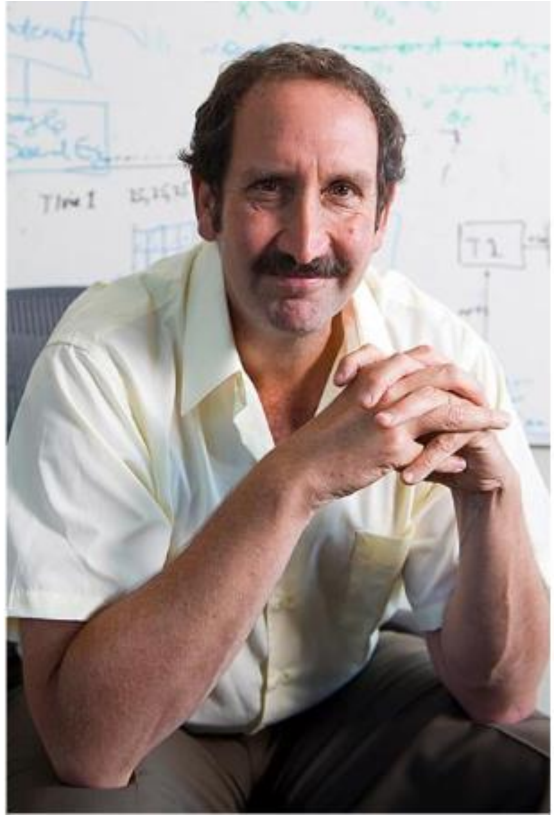


Large Deformation Diffeomorphic Metric Mapping (LDDMM)



Fluid registration





Michael Miller

Computing large deformation metric mappings via geodesic flows of diffeomorphisms. *Int. J. Comput. Vis.* 2005;vol. 61(no. 2):139–139

Large Deformation Diffeomorphic Metric Mapping (LDDMM)

Large Deformation Diffeomorphic Metric Mapping is an image registration algorithm that minimises the following:

$$\mathcal{E} = \frac{1}{2}\int_{t=0}^{1} ||\mathbf{L}\mathbf{v}_t||^2 dt + \frac{1}{2\sigma^2}||f - \mu\left(\varphi_1^{-1}\right)||^2$$

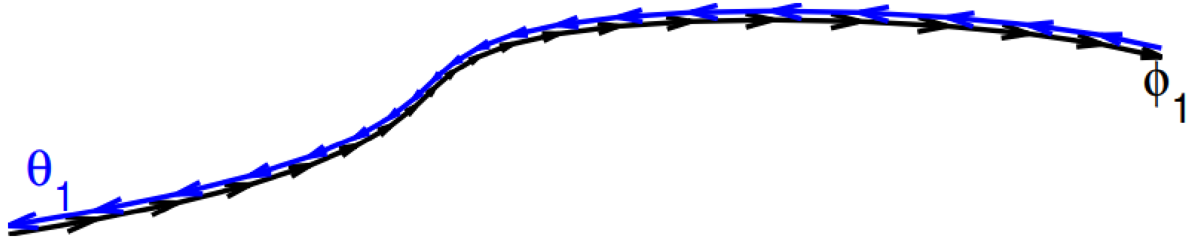
$$\text{where } \varphi_0 = \mathrm{Id},\ \frac{d\varphi}{dt} = \mathbf{v}_t\left(\varphi_t\right)$$

Differential Operator

Large Deformations

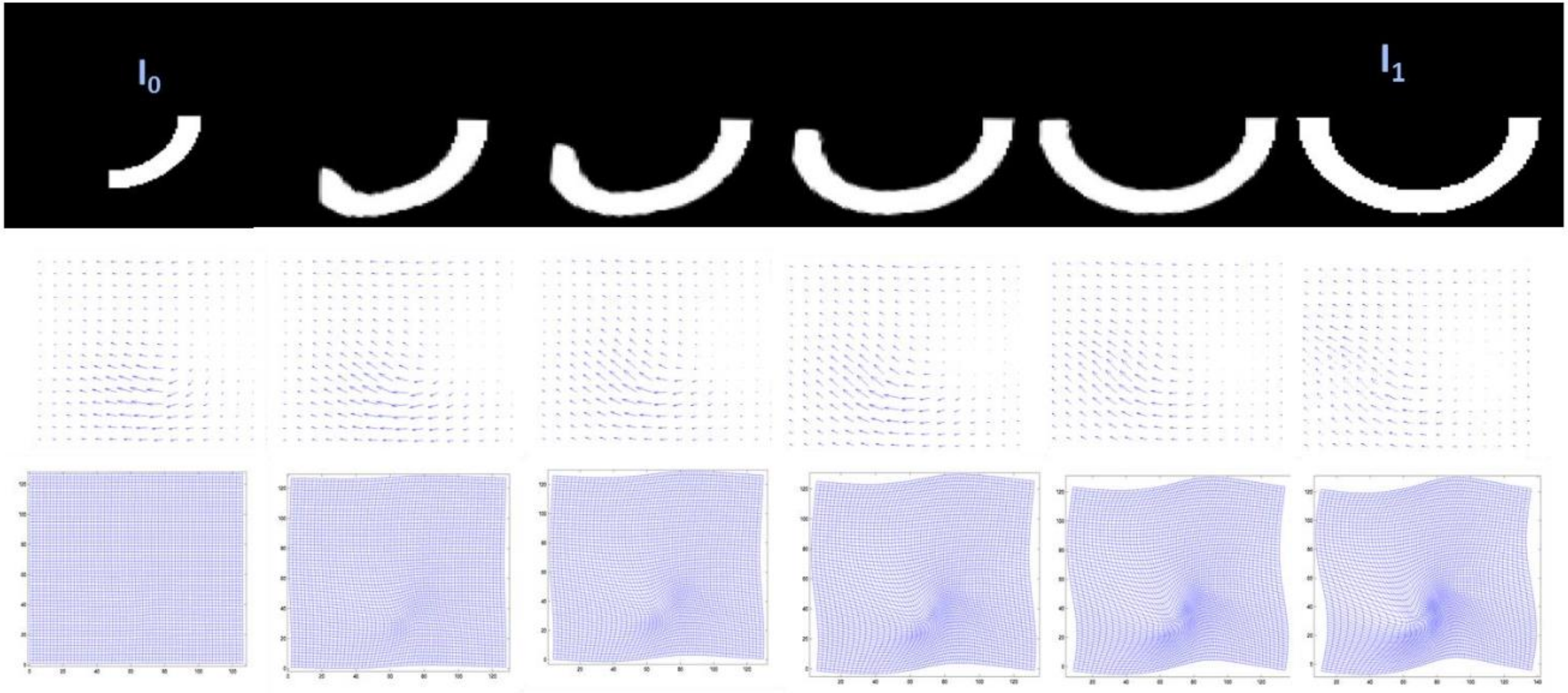
Geodesic Distances

The objective is to estimate a series of velocity fields (v_t)



https://www.fil.ion.ucl.ac.uk/~john/misc/CMIC_ReadingGroup270511.pdf

Example



Jabocian Matrix

- **The Jabocian matrix of transformation:**

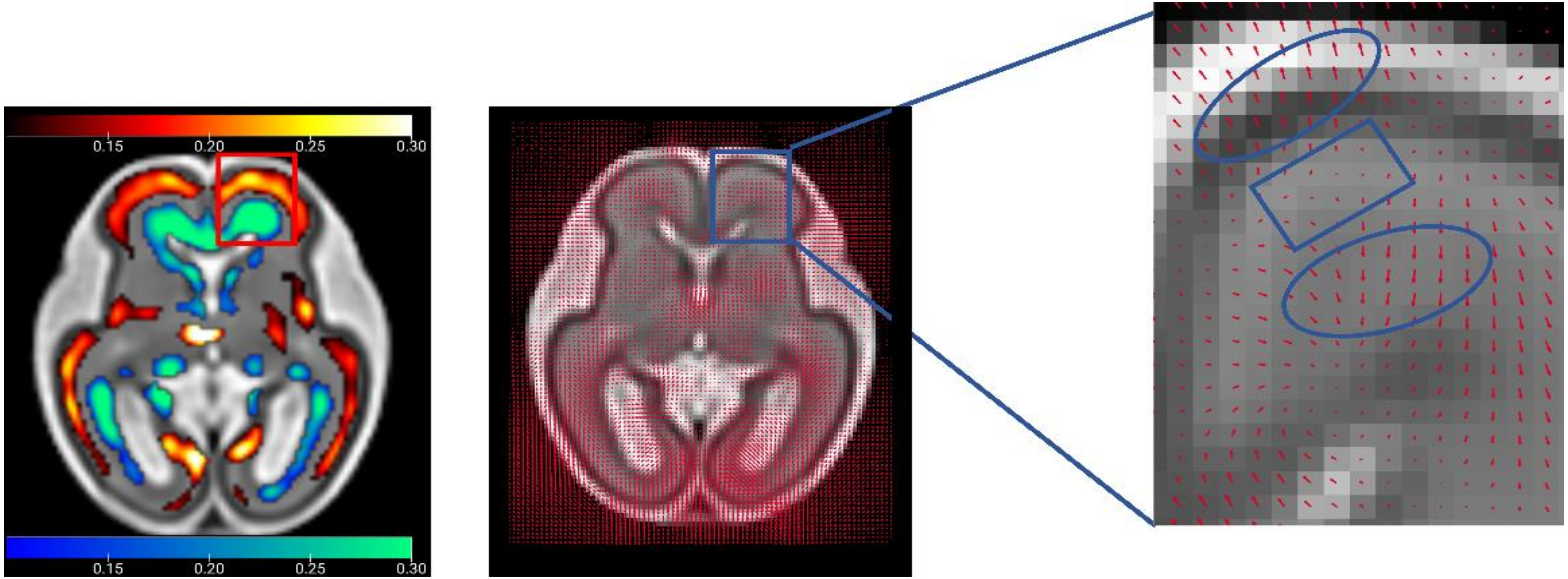
$$J(p) = \begin{bmatrix} \frac{\partial T_x(p)}{\partial x} & \frac{\partial T_x(p)}{\partial y} & \frac{\partial T_x(p)}{\partial z} \\ \frac{\partial T_y(p)}{\partial x} & \frac{\partial T_y(p)}{\partial y} & \frac{\partial T_y(p)}{\partial z} \\ \frac{\partial T_z(p)}{\partial x} & \frac{\partial T_z(p)}{\partial y} & \frac{\partial T_z(p)}{\partial z} \end{bmatrix}$$

- The determinant of this matrix measures how infinitesimal volumes change under the transformation and can be used to analyze the local behavior of the transformation.

$$\boldsymbol{J}(\boldsymbol{p}) = \boldsymbol{det}|\boldsymbol{J}(p)| = \begin{cases} > 1 & volume\ expansion \\ = 1 & no\ volume\ change \\ < 1 & volulme\ contraction \end{cases}$$

Example

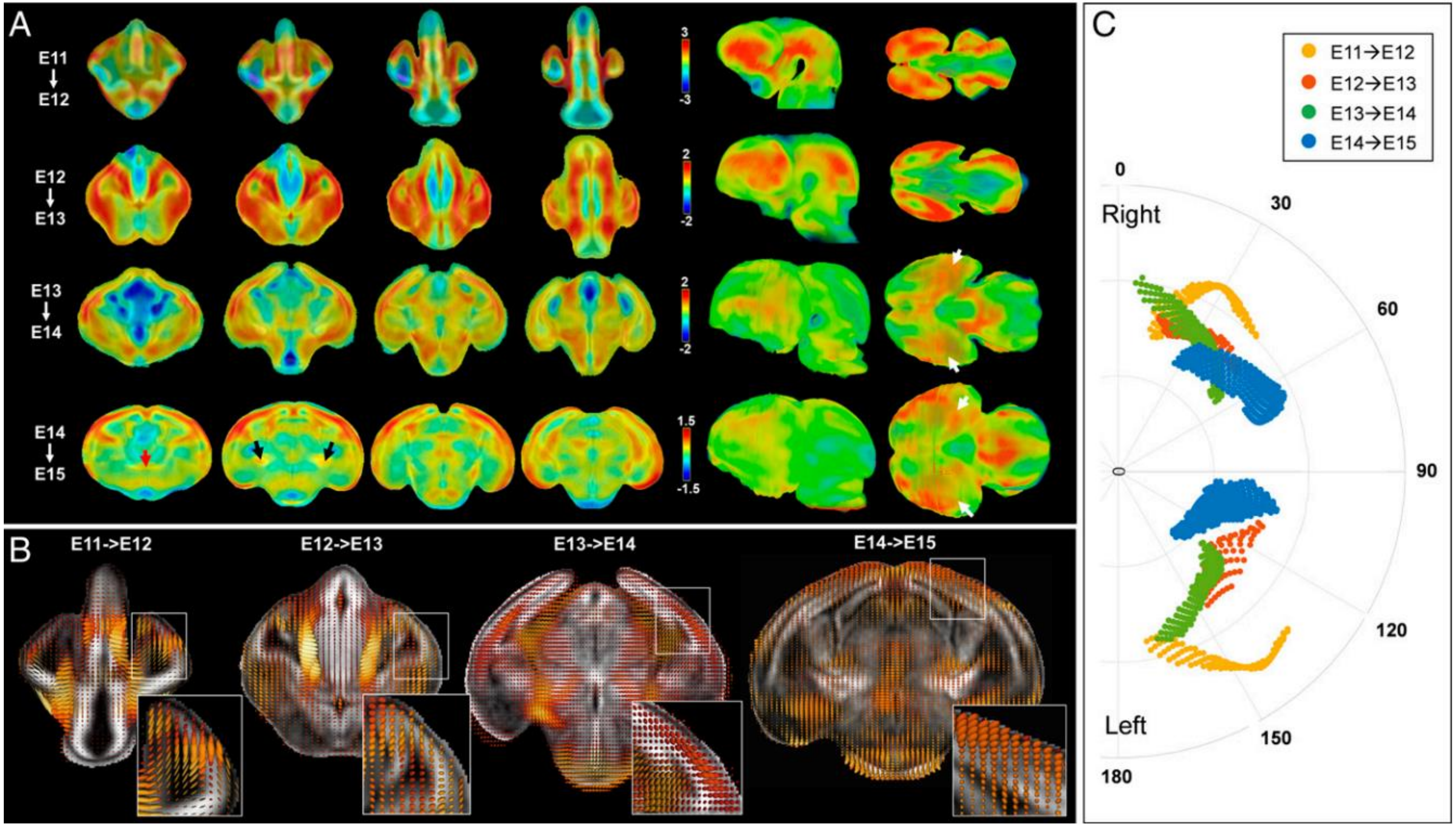
- Tensor based morphometry (TBM)



Determinant (log transformed)

Displacement field

Example



Wu et al., PNAS, 2022

Contents

- Linear registration
  - Previously: interpolation & spatial transform
  - Similarity metric
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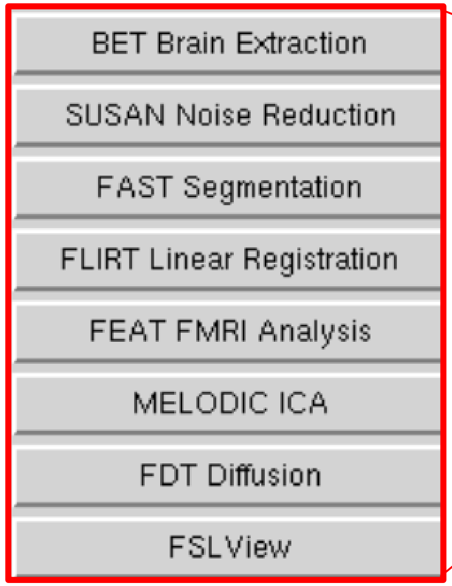
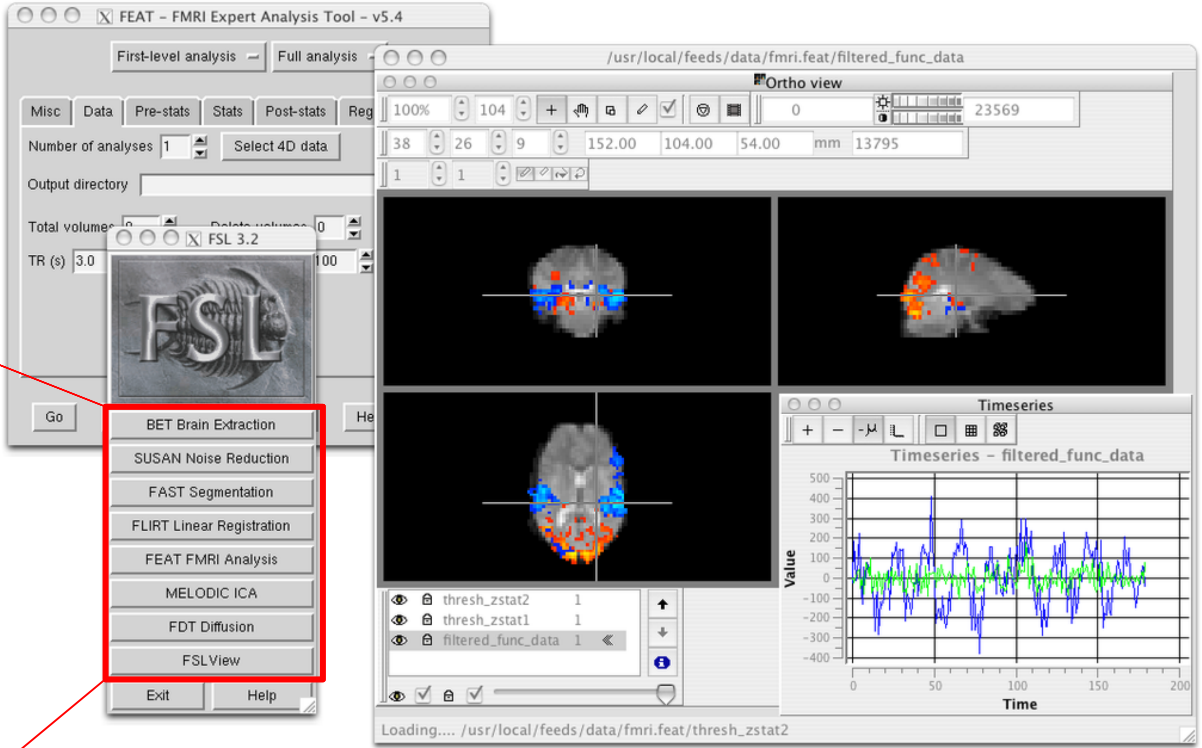
  - Jacobian matrix
- Toolbox and pipeline

- **FSL (FMRIB Software Library)**

A comprehensive library of analysis tools for FMR MRI and DTI brain imaging data.

- Segmentation
- Registration
- Statistics
- ……





- Overview of FSL tools
  - Functional MRI: FEAT, MELODIC, FABBER, BASIL, VERBENA
  - Structural MRI: BET, FAST, FIRST, FLIRT & FNIRT, FSLVBM, SIENA & SIENAX, MIST, BIANCA, MSM, fsl_anat
  - Diffusion MRI: FDT, TBSS, XTRACT, eddy, topup, eddyqc
  - GLM / Stats: GLM general advice, Randomise, PALM, Cluster, FDR, Dual Regression, Mm, FLOBS
  - Other: FSLeyes, Fslutils, Atlases, Atlasquery, SUSAN, FUGUE, MCFLIRT, Miscvis, POSSUM, BayCEST, ICA_PNM, FSL-MRS
  - Using GPUs: Linux GPU requirements

https://fsl.fmrib.ox.ac.uk/fsl/fslwiki

- **FSL – Nonlinear registration**
  - **Finite Element Model-based Registration**

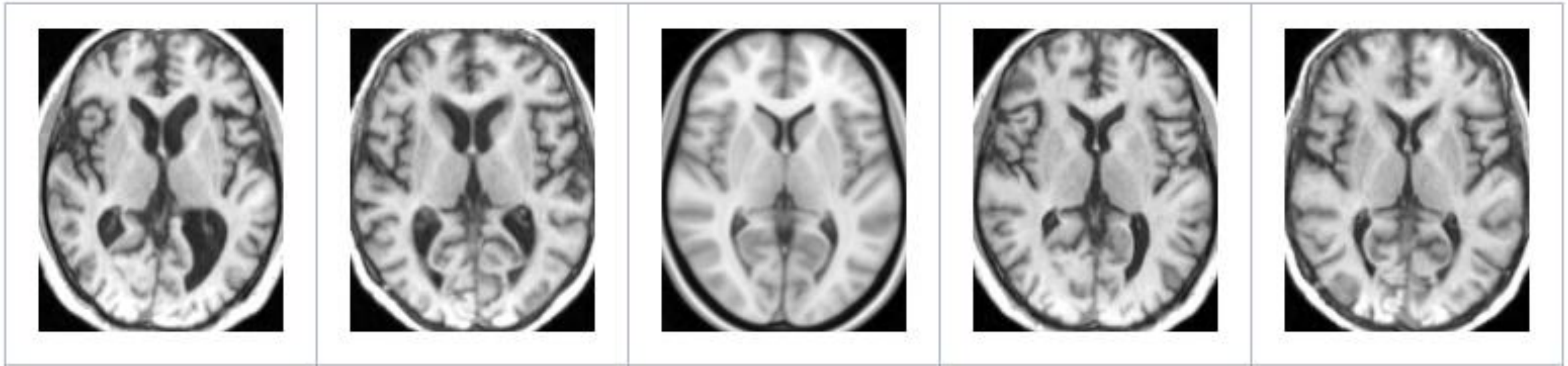
FNIRT

FMRIB's Nonlinear Image Registration Tool

https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/FNIRT

```
fnirt --ref=target_image --in=input_image [options]...
```

```
flirt -ref ${FSLDIR}/data/standard/MNI152_T1_2mm_brain -in my_betted_structural -omat my_affine_transf.mat
fnirt --in=my_structural --aff=my_affine_transf.mat --cout=my_nonlinear_transf --config=T1_2_MNI152_2mm
applywarp --ref=${FSLDIR}/data/standard/MNI152_T1_2mm --in=my_structural --warp=my_nonlinear_transf --out=my_warped_structural
```



flirted subject #1	flirted subject #2	MNI152 T1	fnirted subject #1	fnirted subject #2

- **Freesurfer**

A software package for the analysis and visualization of structural and functional neuroimaging data from cross-sectional or longitudinal studies.

Structural MRI

FreeSurfer provides a full processing stream for structural MRI data, including:

- Skull stripping, B1 bias field correction, and gray-white matter segmentation
- Reconstruction of cortical surface models (gray-white boundary surface and pial surface)
- Labeling of regions on the cortical surface, as well as subcortical brain structures
- Nonlinear registration of the cortical surface of an individual with a stereotaxic atlas
- Statistical analysis of group morphometry differences

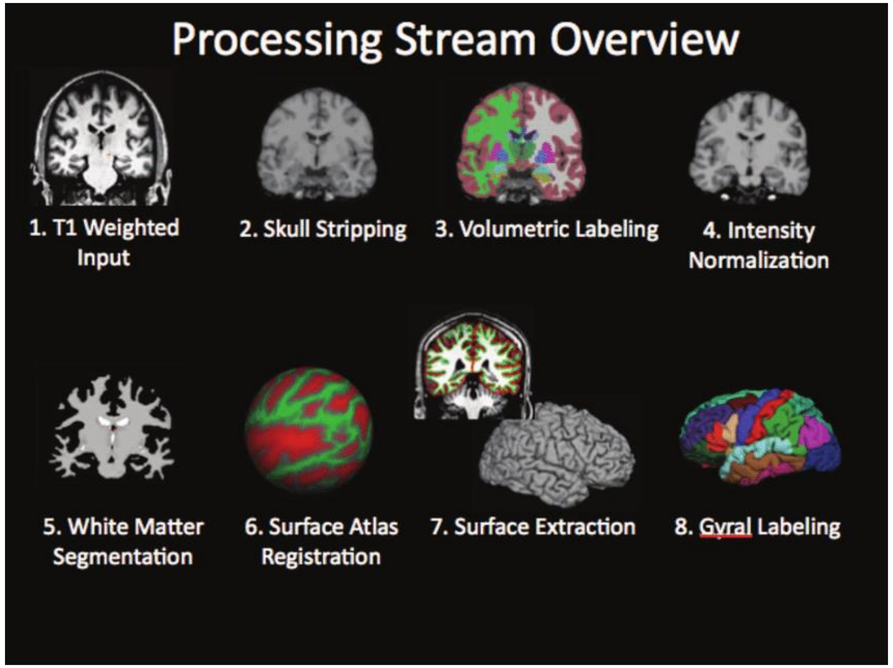
Functional MRI

- FreeSurfer Functional Analysis Stream (FS-FAST):

Diffusion MRI

PET

Multimodal



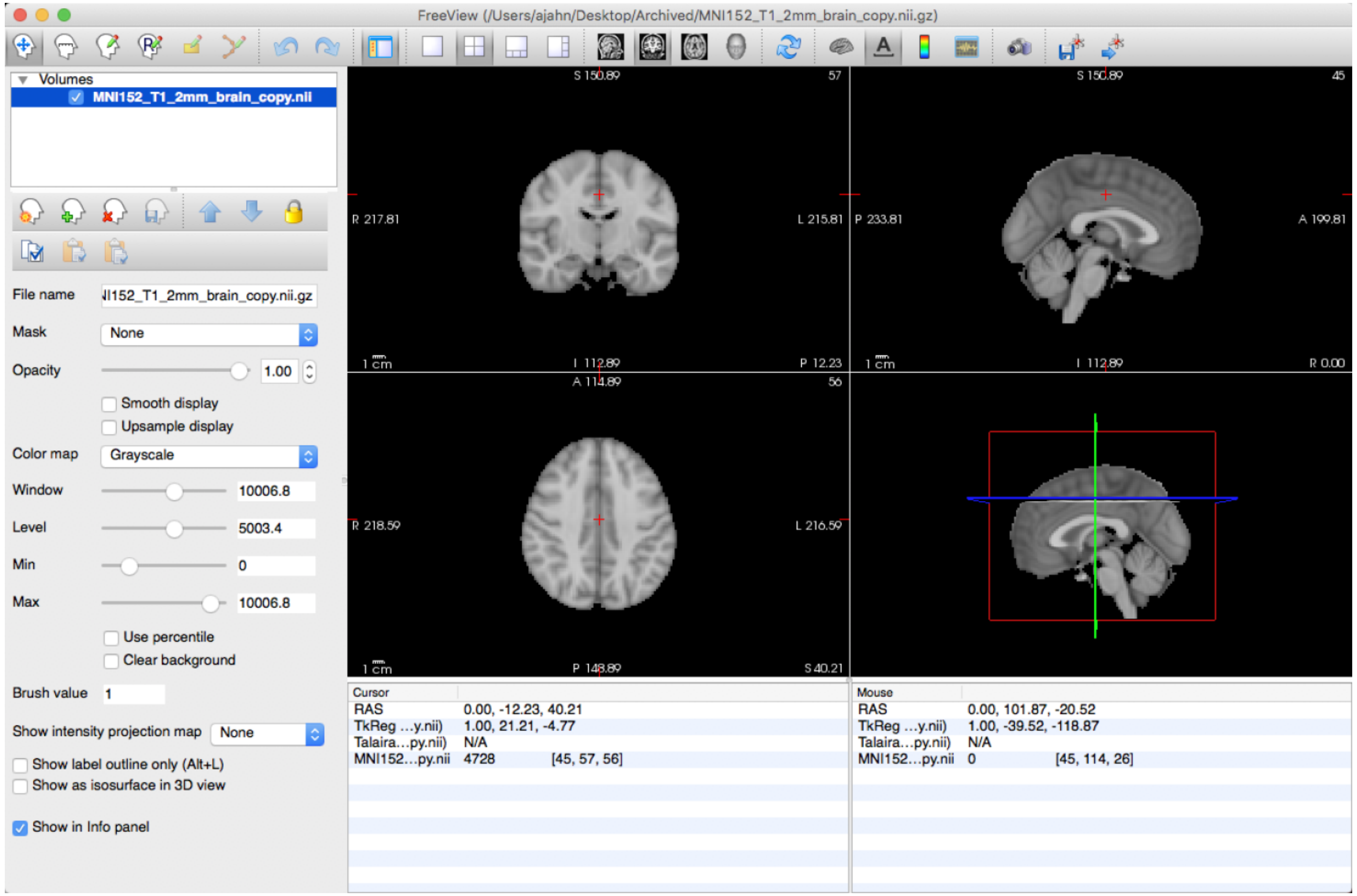
https://surfer.nmr.mgh.harvard.edu/fswiki

• Freesurfer – Nonlinear registration

mri_cvs_register

Performs subject-to-subject or subject-to-atlas non-linear volume registration using a combined volumetric and surface-based (CVS) registration algorithm. It is required that you have an anatomical scan of the subject that has been processed in Freesurfer!

mri_cvs_register --mov <subjid> --template <subjid> **--surf-wm <wm-surface> --surf-cortex <pial-surface>** --no-mask --iscale --c <config-file> ...[option]...



G.M. Postelnicu*, L. Zöllei*, B. Fischl: "Combined Volumetric and Surface Registration", IEEE Transactions on Medical Imaging (TMI), Vol 28 (4), April 2009, p. 508-522

- **Advanced Normalization Tools (ANTs)**

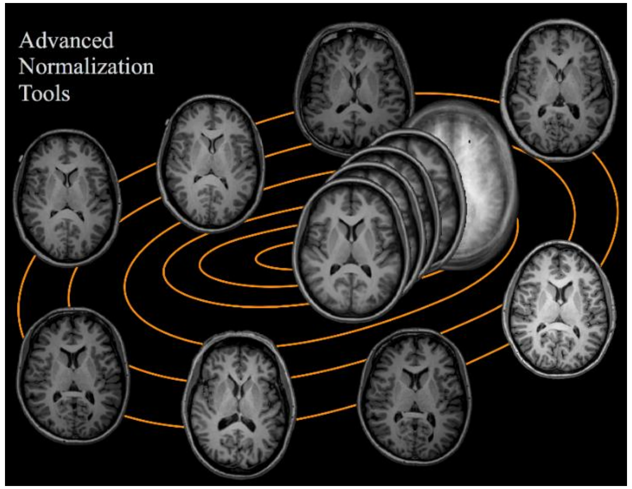
An open-source software package developed for image **registration** and **segmentation** in medical imaging. ANTs provides a comprehensive set of tools and algorithms for advanced image processing and analysis, with a particular focus on **nonlinear registration techniques**.

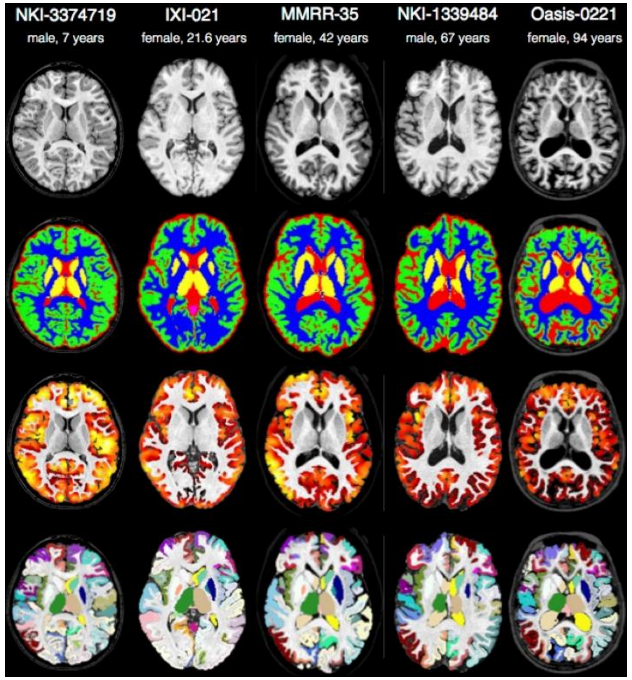
antsRegistration

Rigid, affine, FFD, Diffeomorphic (SyN), etc.

antsRegistration --dimensionality 3 --output [outputPrefix, outputPrefixWarped.nii.gz] --**interpolation** Linear --**use-histogram-matching** 0 **--transform SyN[0.25,3,0]** **--metric MI**[fixedImage.nii.gz,movingImage.nii.gz,1,32,Regular,0.25] --convergence [100x70x50x20,1e-6,10] --shrink-factors 8x4x2x1 --smoothing-sigmas 3x2x1x0vox

-t Rigid; Affine; Translation; BsplineDisplacementField; TimeVaryingvelocityField; SyN [gradientstep]

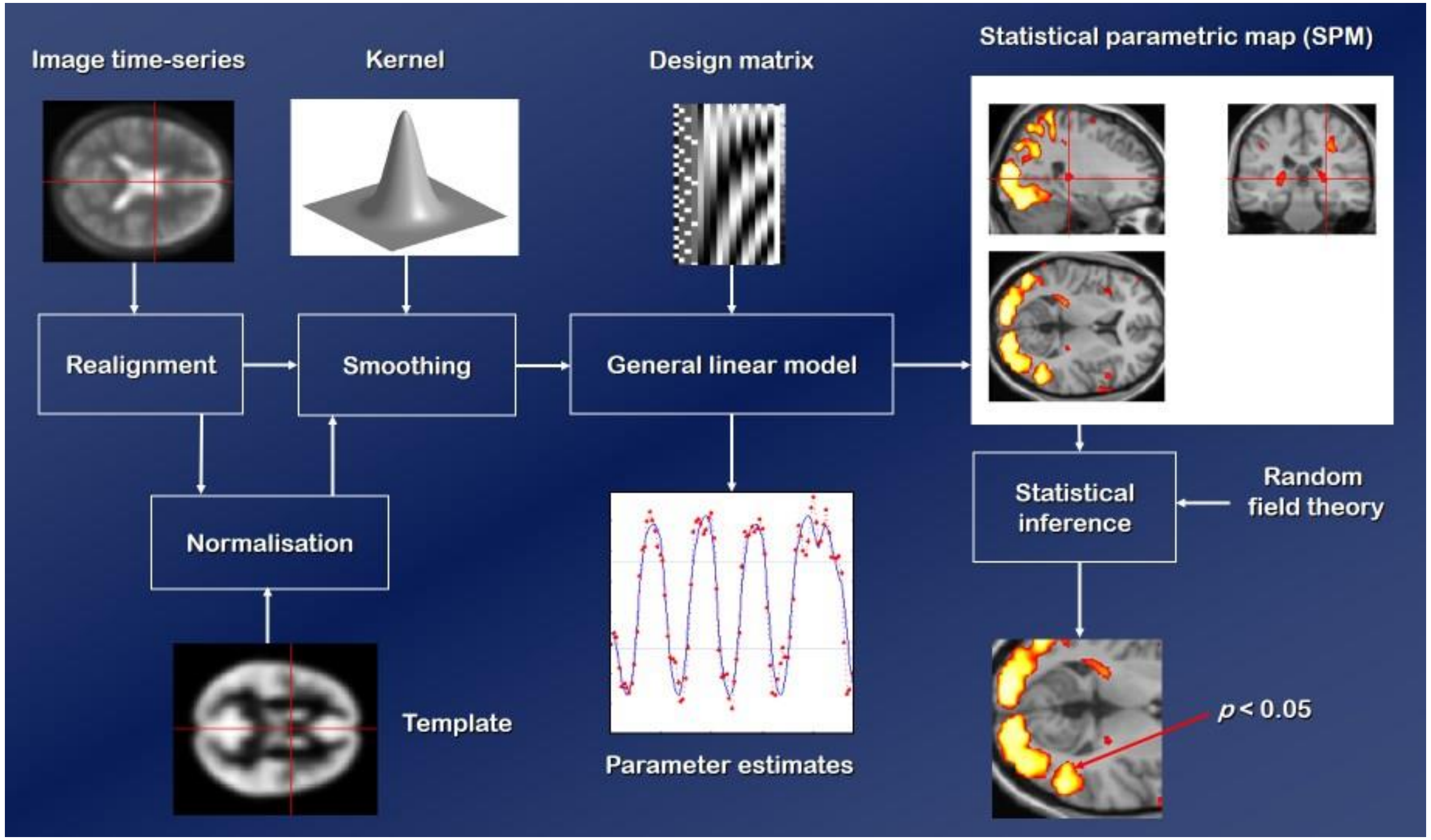




https://github.com/ANTsX/ANTs

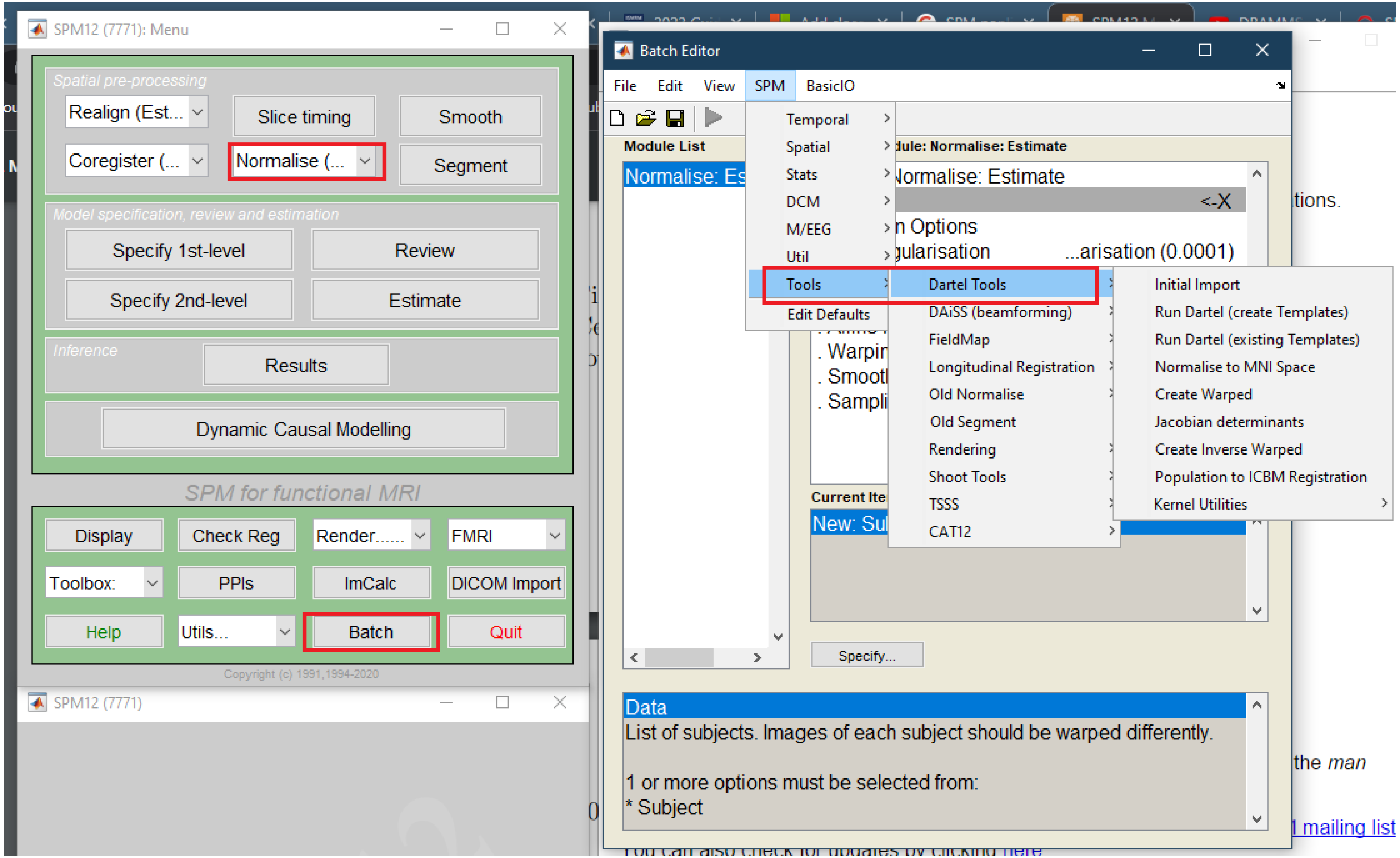
- **Statistical Parametric Mapping (SPM)**

Statistical Parametric Mapping refers to the construction and assessment of spatially extended statistical processes used to test hypotheses about functional imaging data.



- **Spatial processing**
  - **Registration**
  - Segmentation
- fMRI statistics
- EEG/MEG

- **SPM – DARTEL (Diffeomorphic Anatomical Registration Through Exponentiated Lie Algebra)**



Summary

- Linear registration
  - Interpolation
  - Spatial transform
  - Similarity metric
  - Optimization
- Non-linear registration
  - Diffeomorphic transformation
  - Jacobian matrix
- Toolbox and pipeline

Homework 1

- For two MRI A (fixed) and B' (moving), if two images are same subject scanned for two times, so only linear transform exists between A and B', which can represent as: $B'(i) = kA(i) + t, \forall i \in \{m|B'(m) > 0\}$.

- Prove that:
  - The CC between A and B' is 1.

  - Represent the SSD between A and B' using k, t, total pixel number N and the mean/std value of A ($\bar{A}$, $\sigma(A)$).

Homework 2

- For image $A = [18, 97]$, and $B = [97, 18]$.

1. Calculate the $H(A)$ and $H(B)$
   (save the result with $\log(m)\,, m \in N$, e.g. $3\log(5) + 2\log(6)$)

2. Calculate the $I(A, B)$ and $NMI(A, B) = \frac{H(A)+H(B)}{H(A,B)}$

3. If the image add more background: $A' = [18, 97, \overbrace{0, 0, \ldots, 0}^{N}]$ and $B = [97, 18, 0, 0, \ldots, 0]$, calculate the $I(A', B')$ and $NMI(A', B')$, discuss why NMI is more frequency used?

Homework3

- Suppose the spatial transformation is represented as:

$$\begin{cases} x' = \sin(\mathrm{x} - 2\mathrm{y}) \\ y' = \cos(\mathrm{x} + \mathrm{y}) \end{cases}$$

- Calculate the Jabocian matrix;
- Calculate the Jabocian determinant;
- Discuss the volume change (expansion/contraction/no change) for coordinate $(x, y) = \left(\frac{\pi}{6}, \frac{\pi}{2}\right), \left(0, \frac{7\pi}{6}\right), (\frac{5\pi}{6}, 0)$